

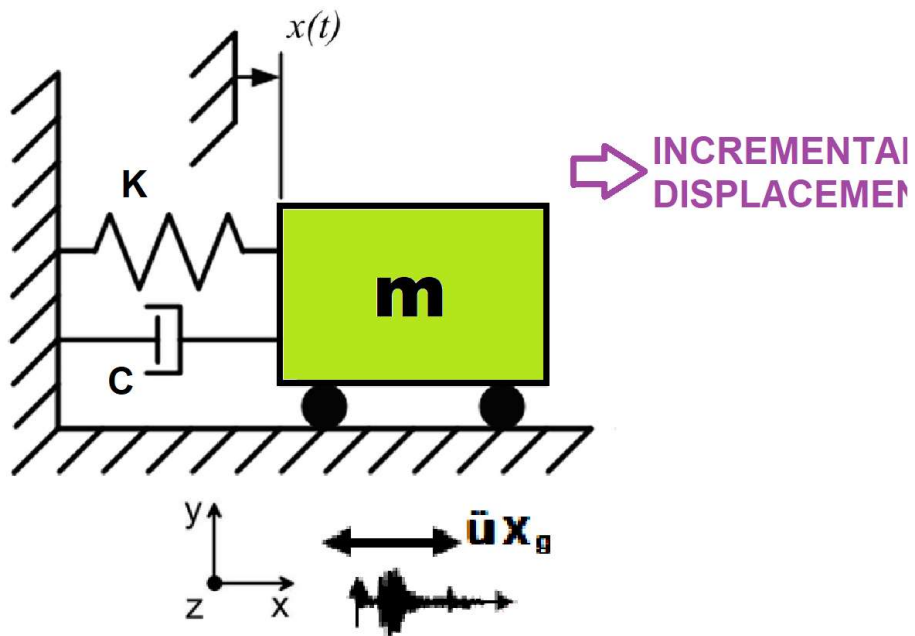
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS OF STRUCTURE DUCTILITY RATIO AND OPENSEES VIA 20 SEISMIC GROUND MOTIONS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

Assume that a single-degree-of-freedom structure is subjected to nonlinear dynamic analysis under twenty different ground motion records. To investigate the influence of structure ductility on seismic performance, a sensitivity analysis is performed by systematically varying each column's ductility ratio from 5.0 to 15.0 in 21 steps. For each ductility level, a pushover analysis provides a cyclic displacement analysis establishes a reference hysteretic energy, and twenty nonlinear time-history analyses are conducted using different ground motions. From these analyses, the Energy Dissipation Capacity Index (EDCI) and other response metrics such as maximum displacement, velocity, acceleration, damage index, over-strength factor, ductility ratio, and equivalent viscous damping ratio are computed.

The median response over the twenty ground motions is then evaluated for each ductility level. Finally, the results are assessed through trend plots, 3D contour surfaces, correlation heatmaps, Random Forest regression, and ANOVA sensitivity analysis to identify the optimal column ductility ratio that maximizes energy dissipation capacity while maintaining structural safety.



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

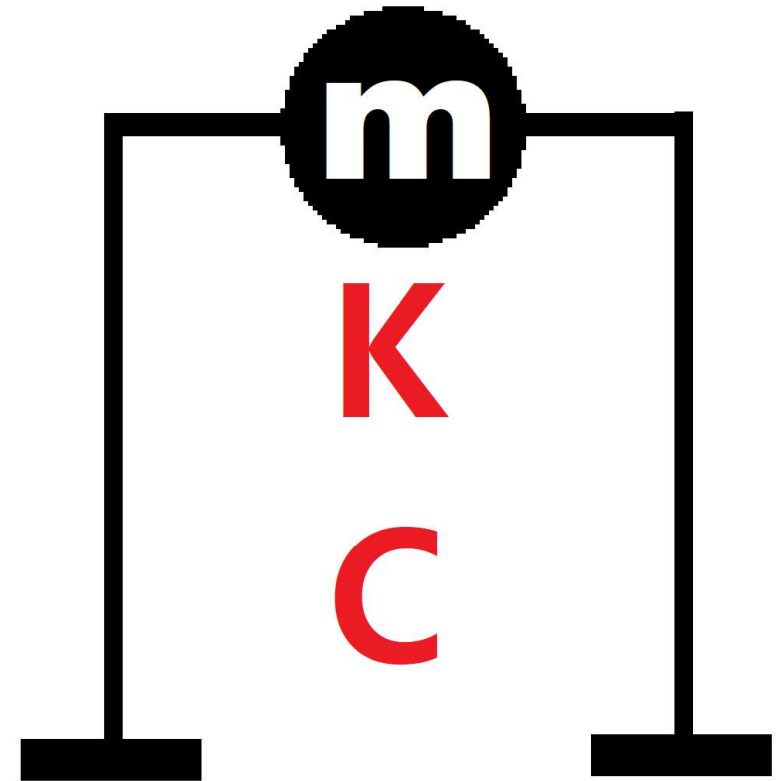
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

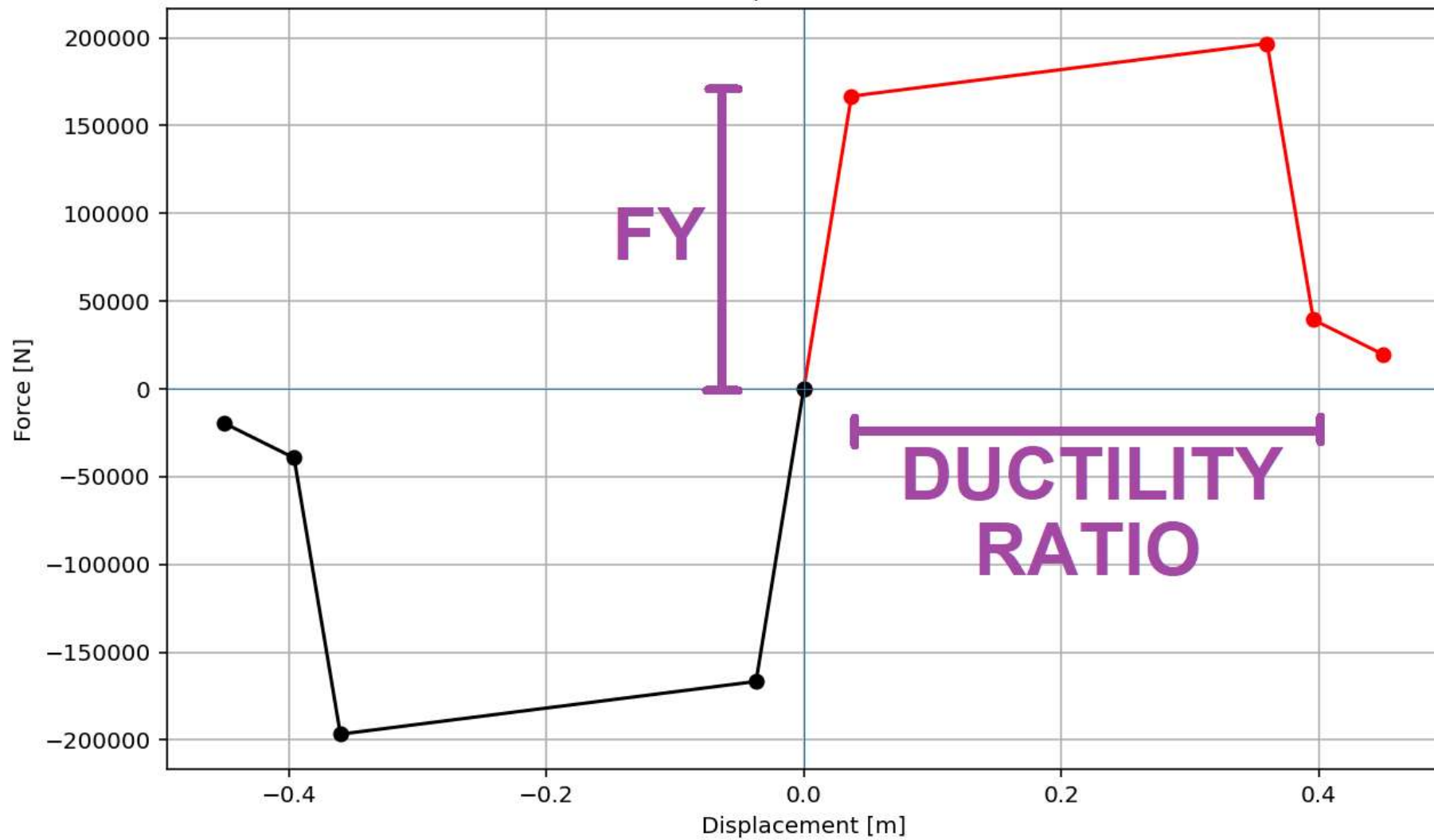
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



Force-Displacement Curve



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DUCT_SENSITIVITY_20_SEISMIC_SDOF.py X

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3#>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

4#SENSITIVITY ANALYSIS OF STRUCTURAL DUCTILITY RATIO AND OPENSEES VIA 20 SEISMIC GROUND MOTIONS

5#

6#EQUIVALENT VISCOUS DAMPING RATIO: $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$

7#

8#ENERGY DISSIPATION CAPACITY INDEX = $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$

9#

10# $P(t) = P_0 \sin(\omega t)$

11# $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$

12#

13#ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION

14#OF ENERGY DISSIPATION CAPACITY INDEX

15#

16#THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

17#EMAIL: salar.d.ghashghaei@gmail.com

18#####

19Assume that a single-degree-of-freedom structure is subjected to nonlinear dynamic analysis under

20twenty different ground motion records. To investigate the influence of column ductility on

21seismic performance, a sensitivity analysis is performed by systematically varying each column's

22ductility ratio from 20 to 50 in 21 steps. For each ductility level, a pushover analysis provides a c)

23The median response over the twenty ground motions is then evaluated for each ductility level.

24Finally, the results are assessed through trend plots, 3D contour surfaces, correlation heatmaps,

25Random Forest regression, and ANOVA sensitivity analysis to identify the optimal column ductility

26ratio that maximizes energy dissipation capacity while maintaining structural safety.

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30Nonlinear Seismic Performance Assessment of a MDOF:

31An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake L

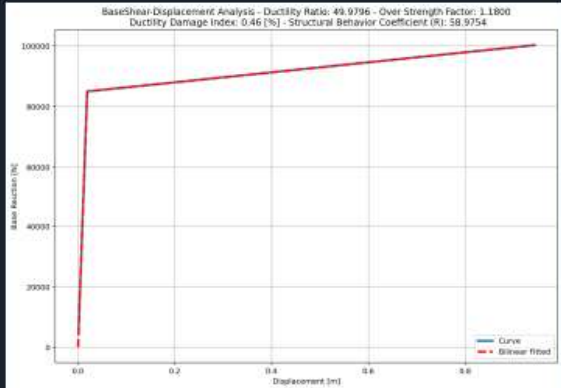
32

33This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF

34for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity

24 %

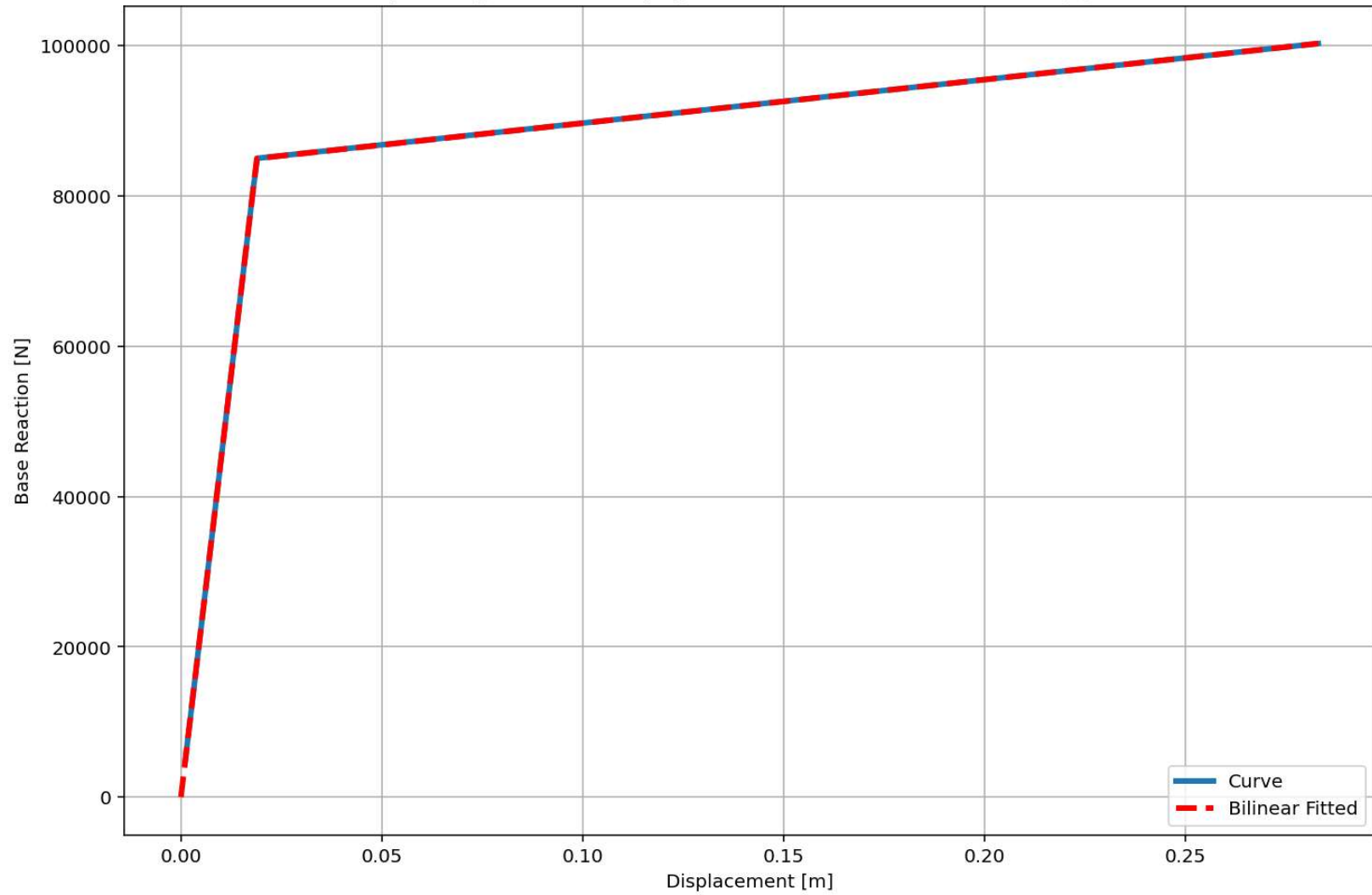
BaseShear-Displacement Analysis - Ductility Ratio: 49.9796 - Over Strength Factor: 1.1800
Ductility Damage Index: 0.46 (%) - Structural Behavior Coefficient (R): 58.9754



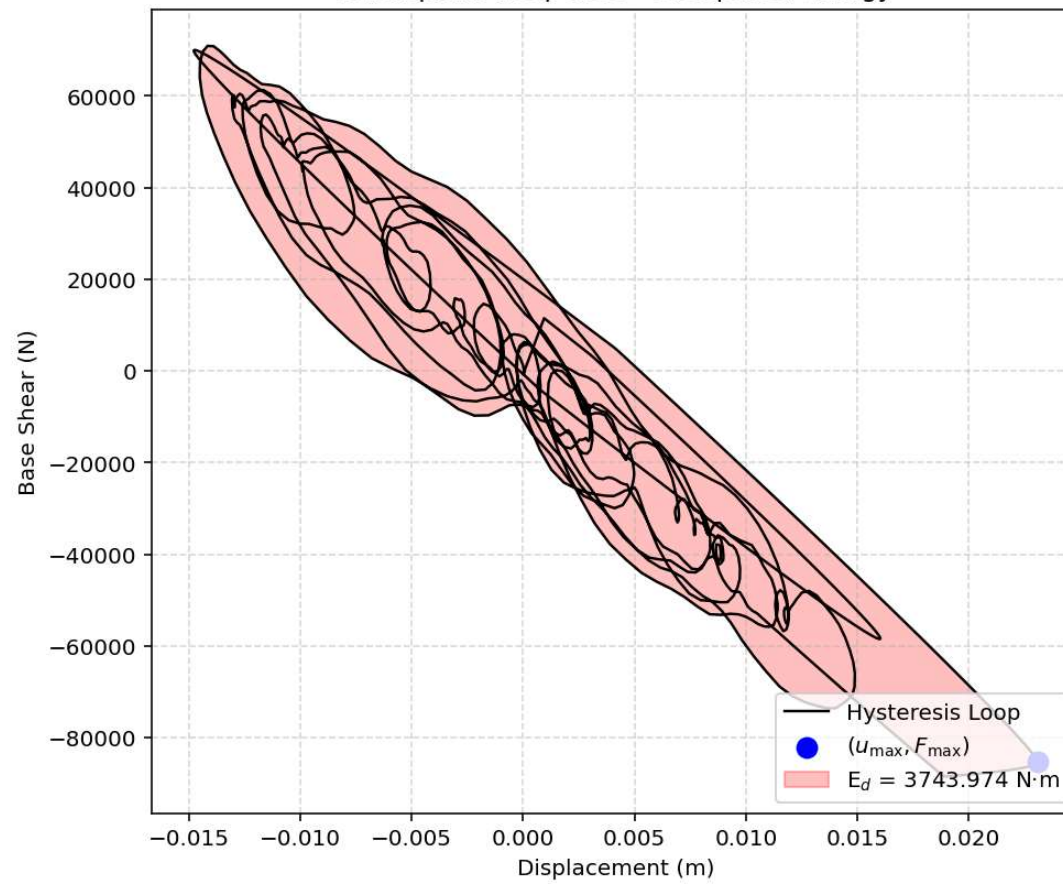
IPython ConsolePlotsVariable ExplorerHelpDebuggerHistoryFiles

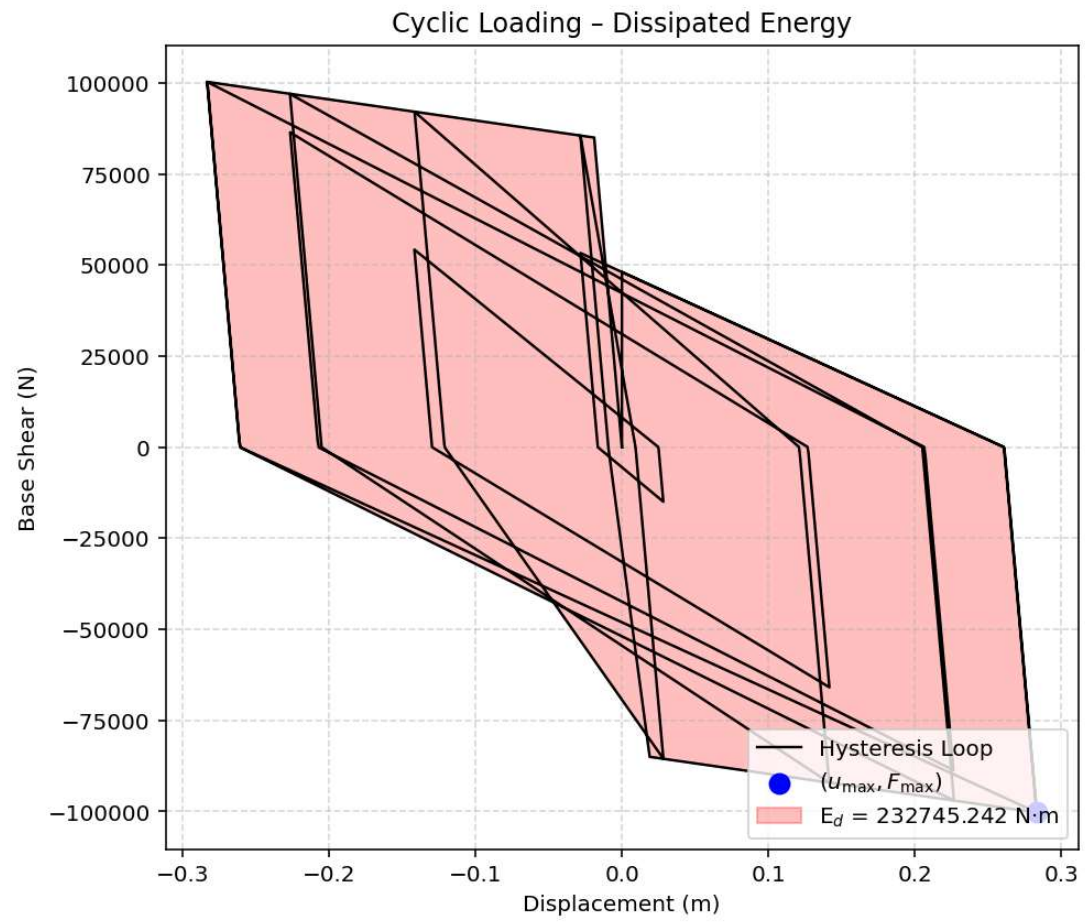
InlineConda: anaconda3 (Python 3.12.7) ✓ LSP: PythonLine 27, Col 1UTF-8CRLF-RWMem 97%

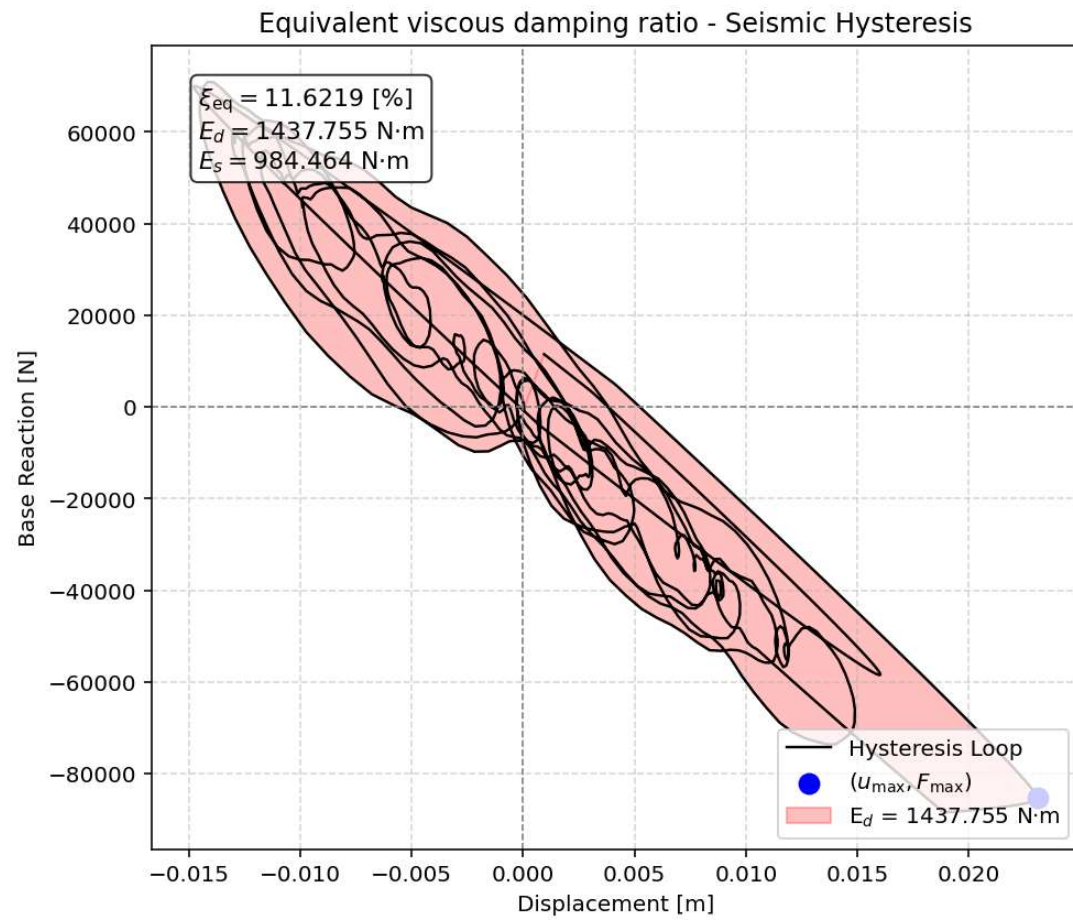
BaseShear-Displacement Analysis - Ductility Ratio: 14.9857 - Over Strength Factor: 1.1800
Ductility Damage Index: 1.60 [%] - Structural Behavior Coefficient (R): 17.6837



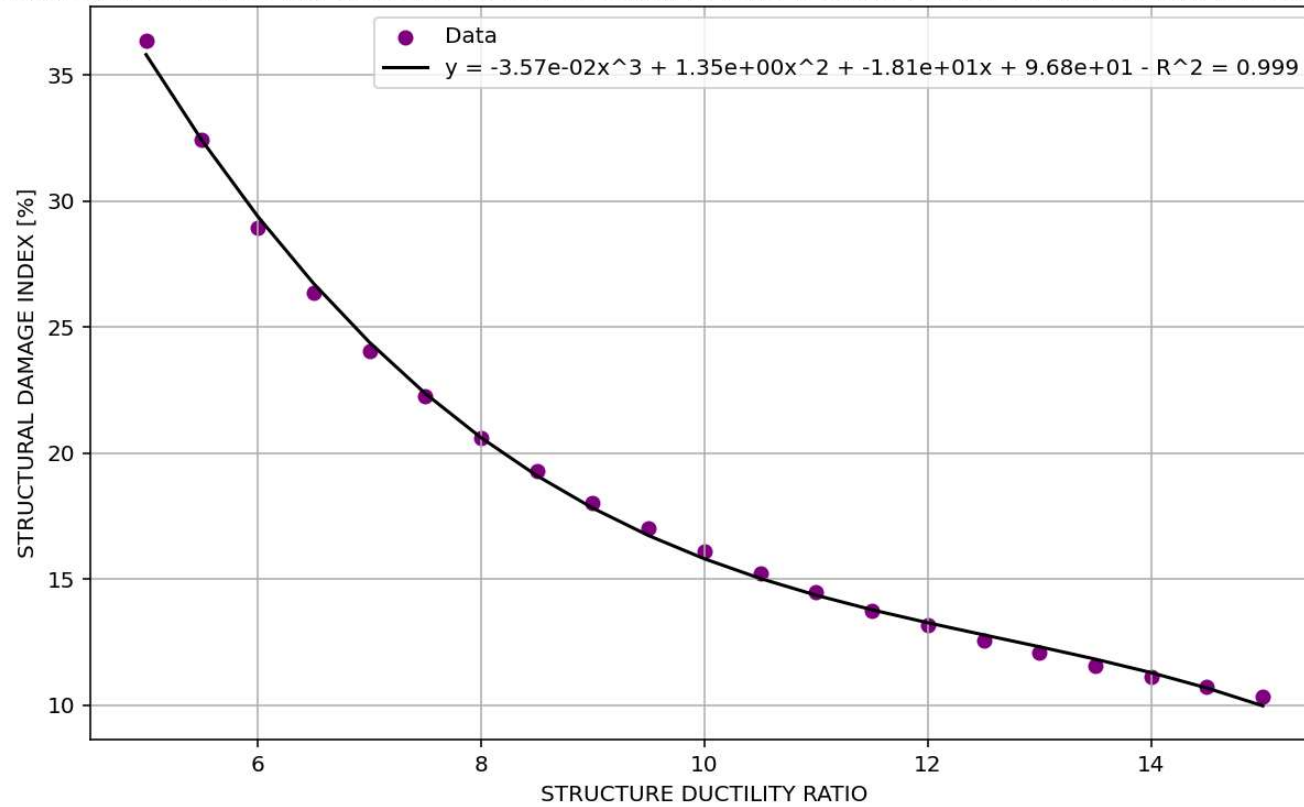
Earthquake Response - Dissipated Energy



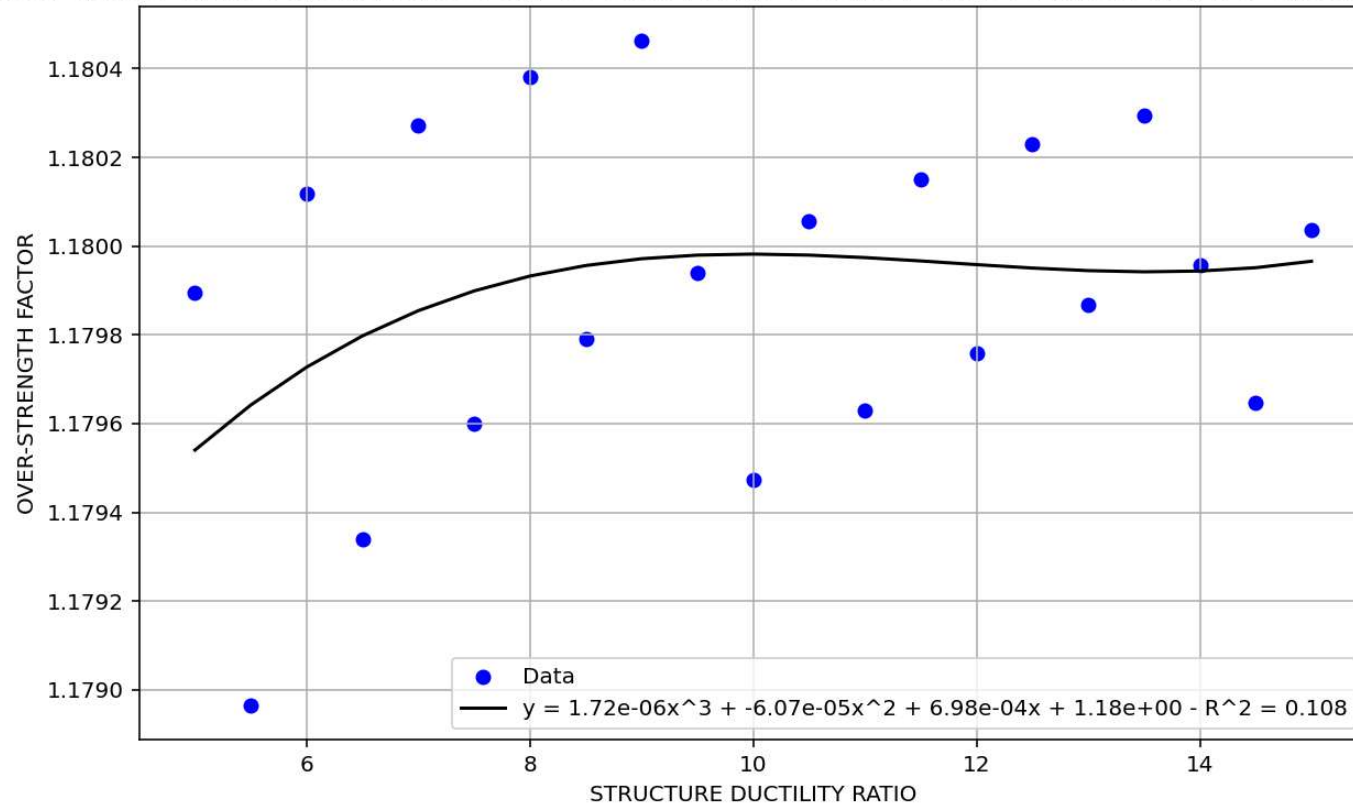




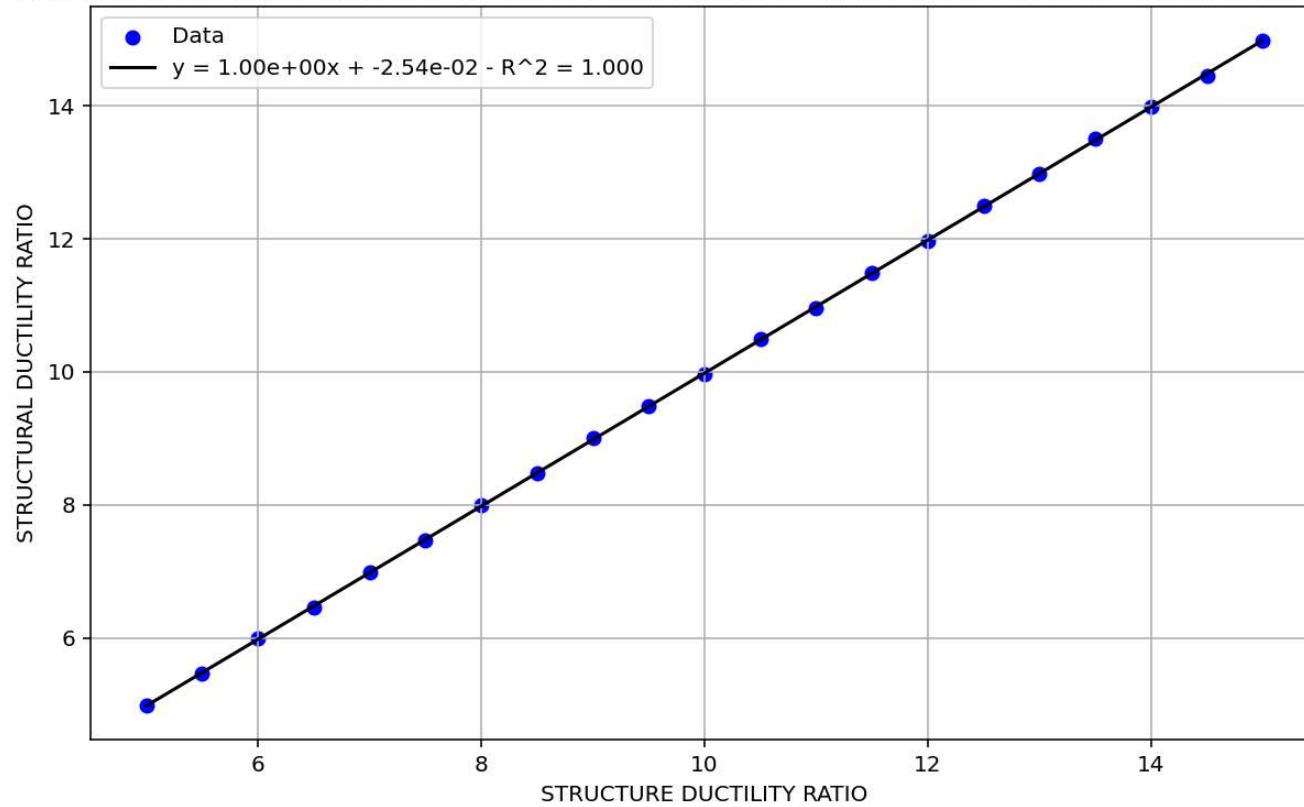
STRUCTURAL DAMAGE INDEX [%] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



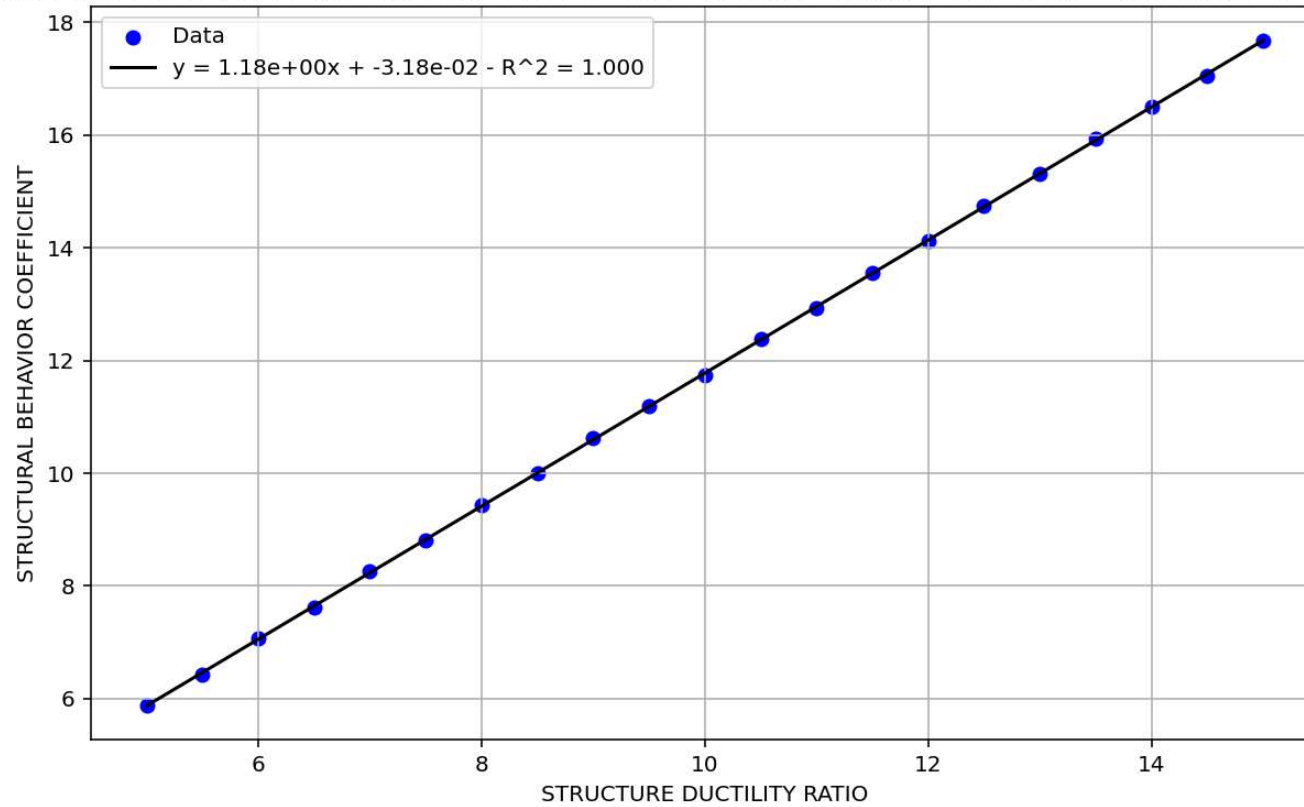
OVER-STRENGTH FACTOR and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



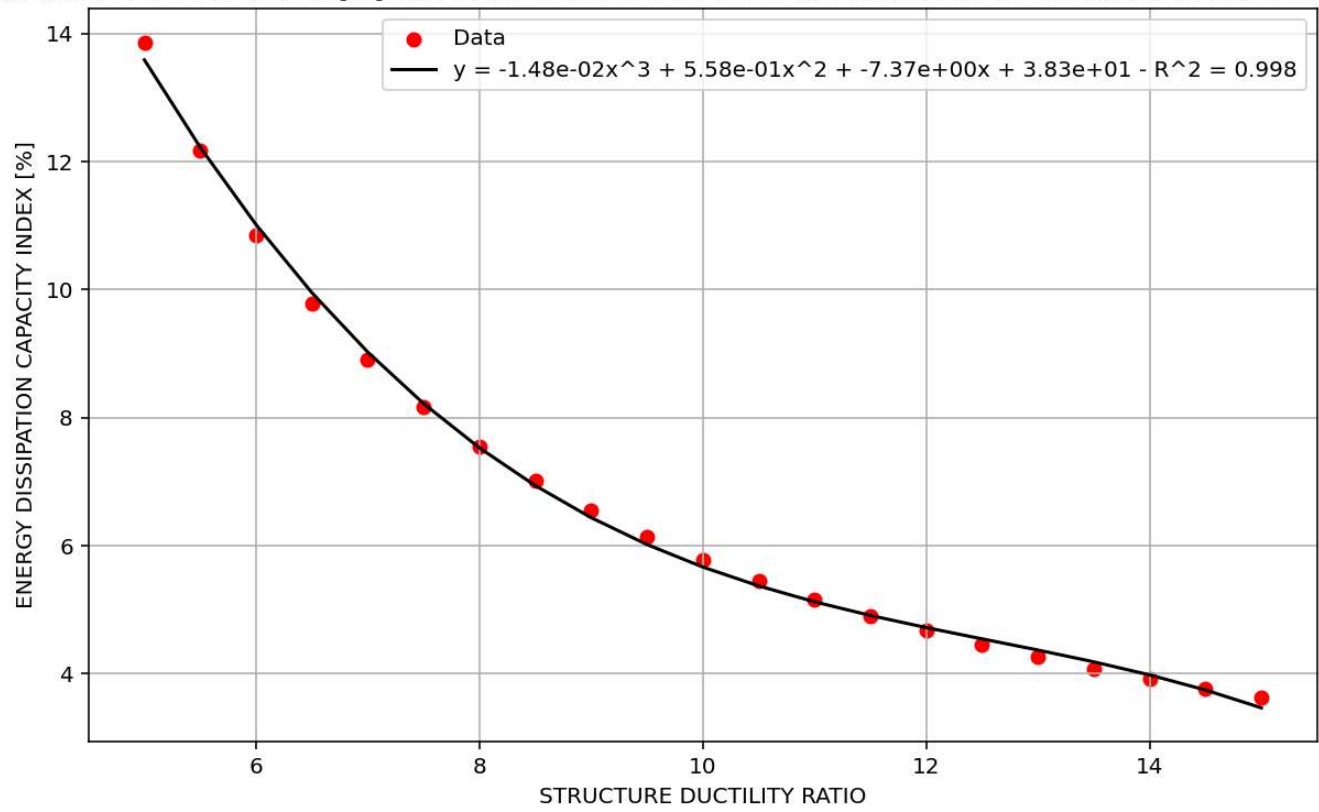
STRUCTURAL DUCTILITY RATIO and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



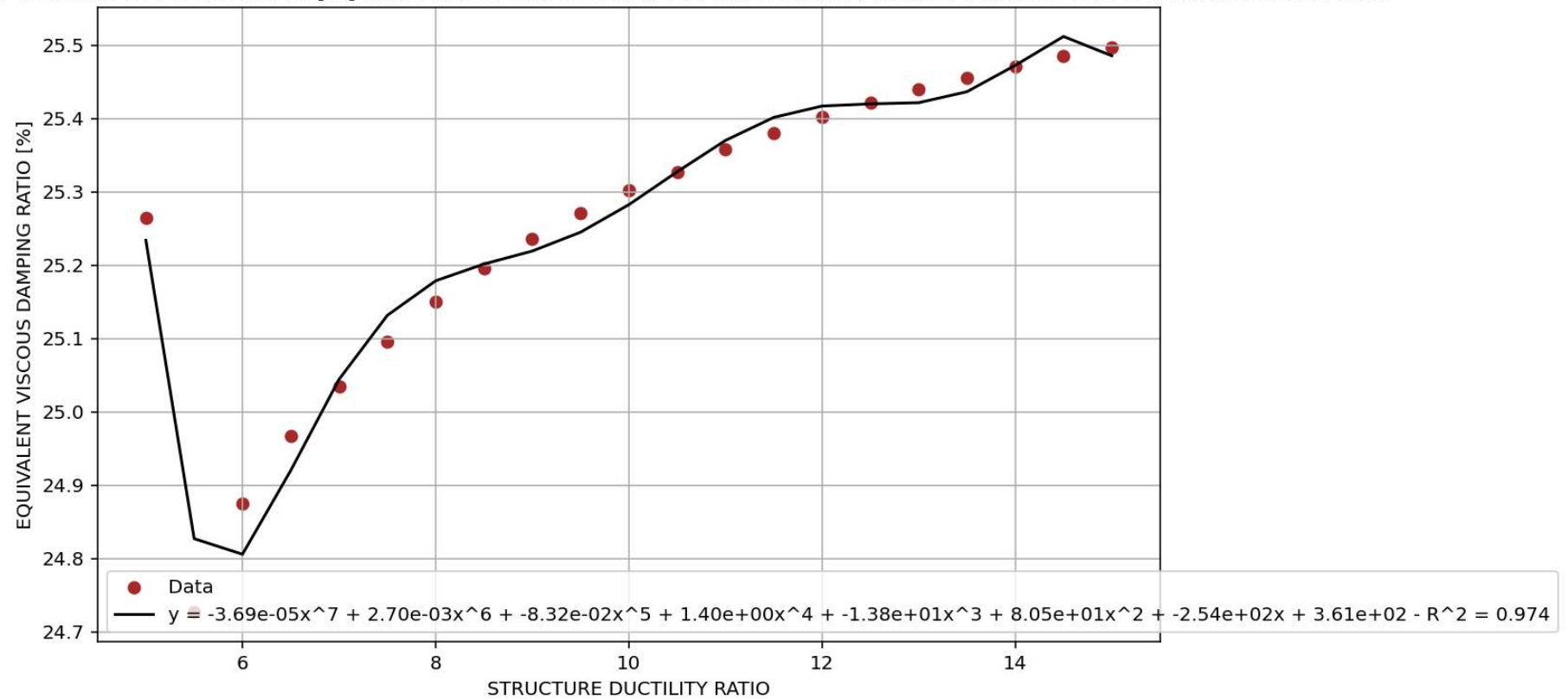
STRUCTURAL BEHAVIOR COEFFICIENT and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



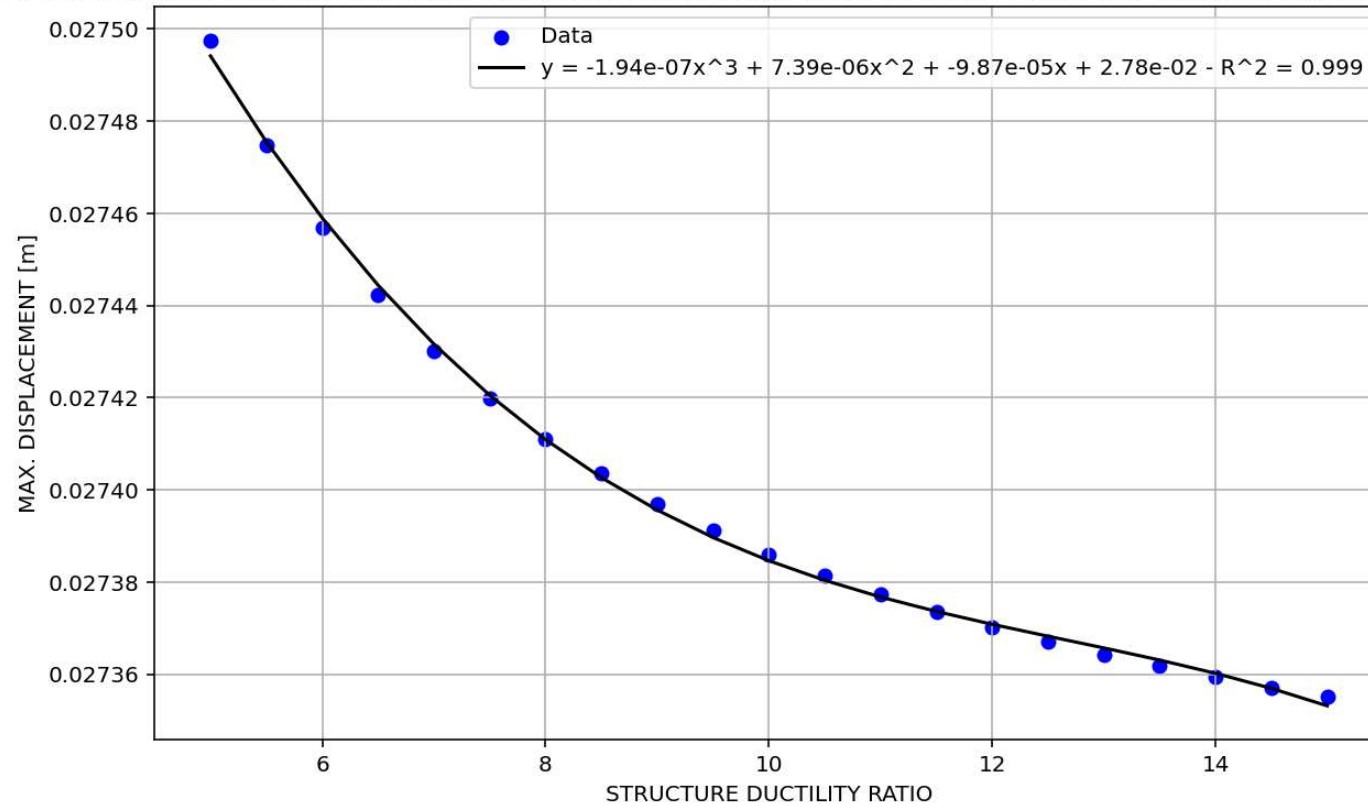
ENERGY DISSIPATION CAPACITY INDEX [%] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



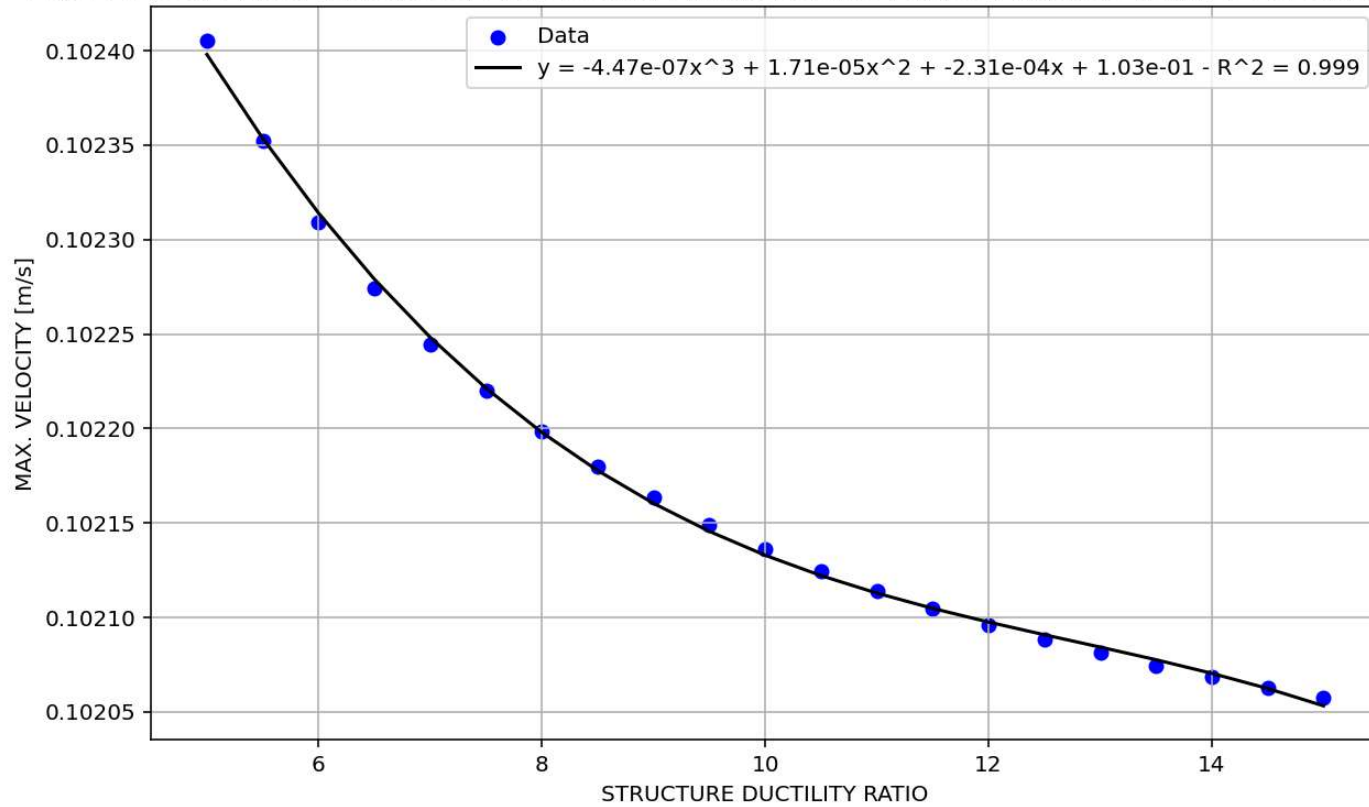
EQUIVALENT VISCOUS DAMPING RATIO [%] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



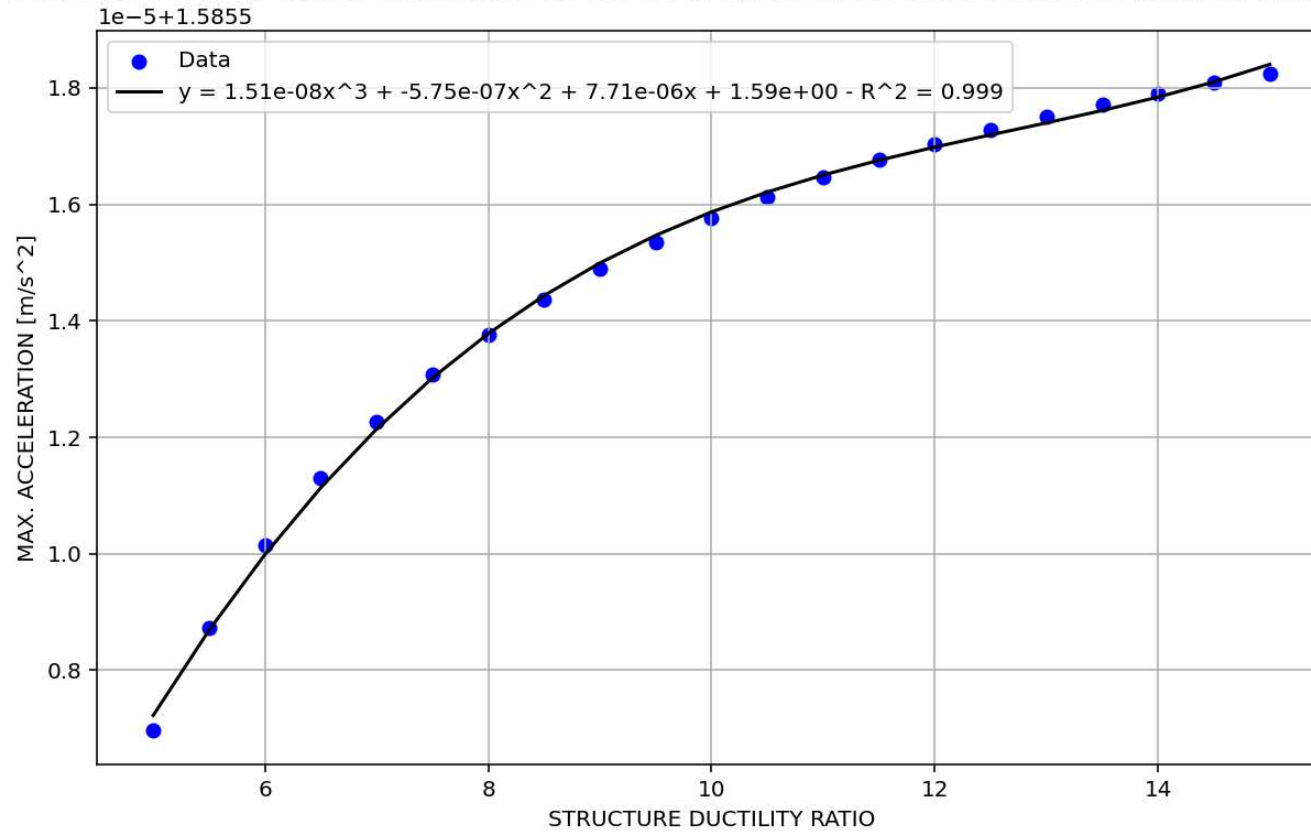
MAX. DISPLACEMENT [m] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



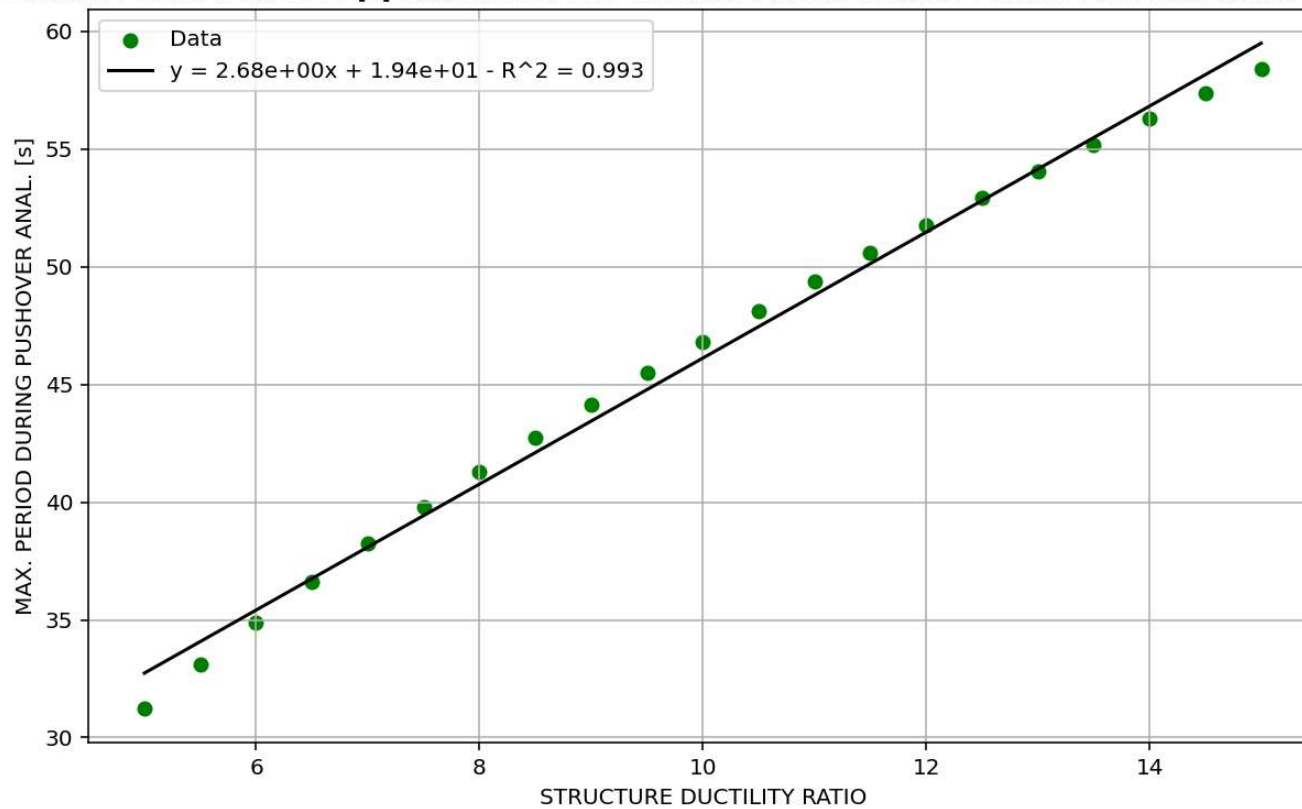
MAX. VELOCITY [m/s] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



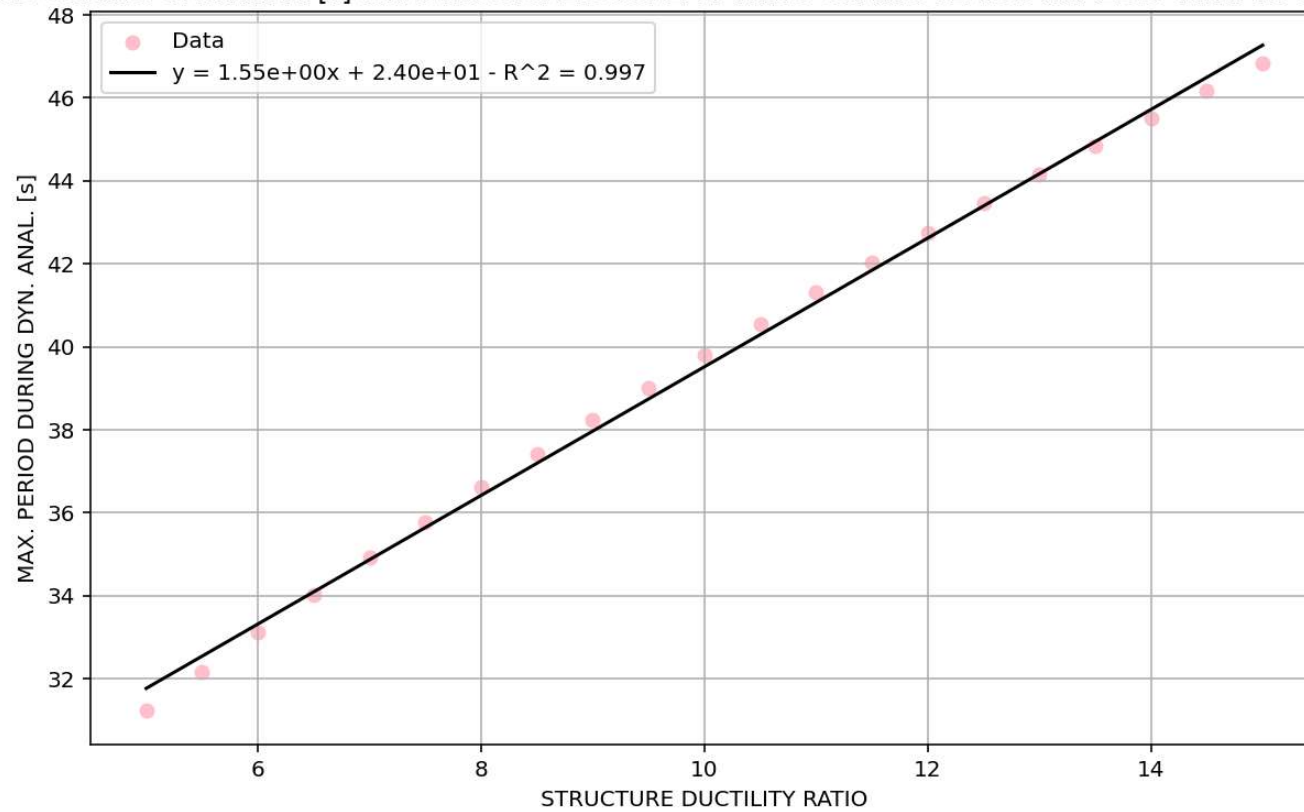
MAX. ACCELERATION [m/s²] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



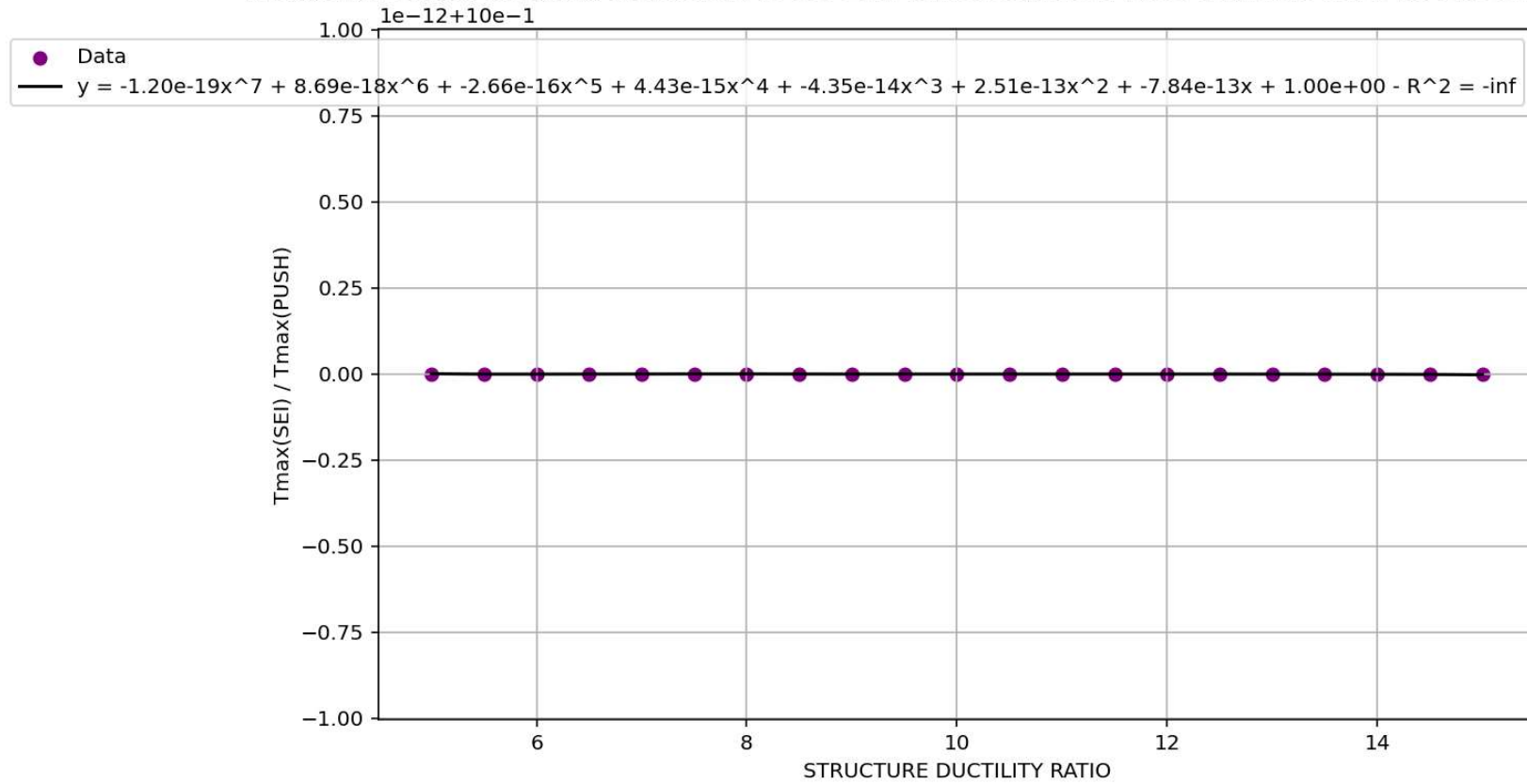
MAX. PERIOD DURING PUSHOVER ANAL. [s] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



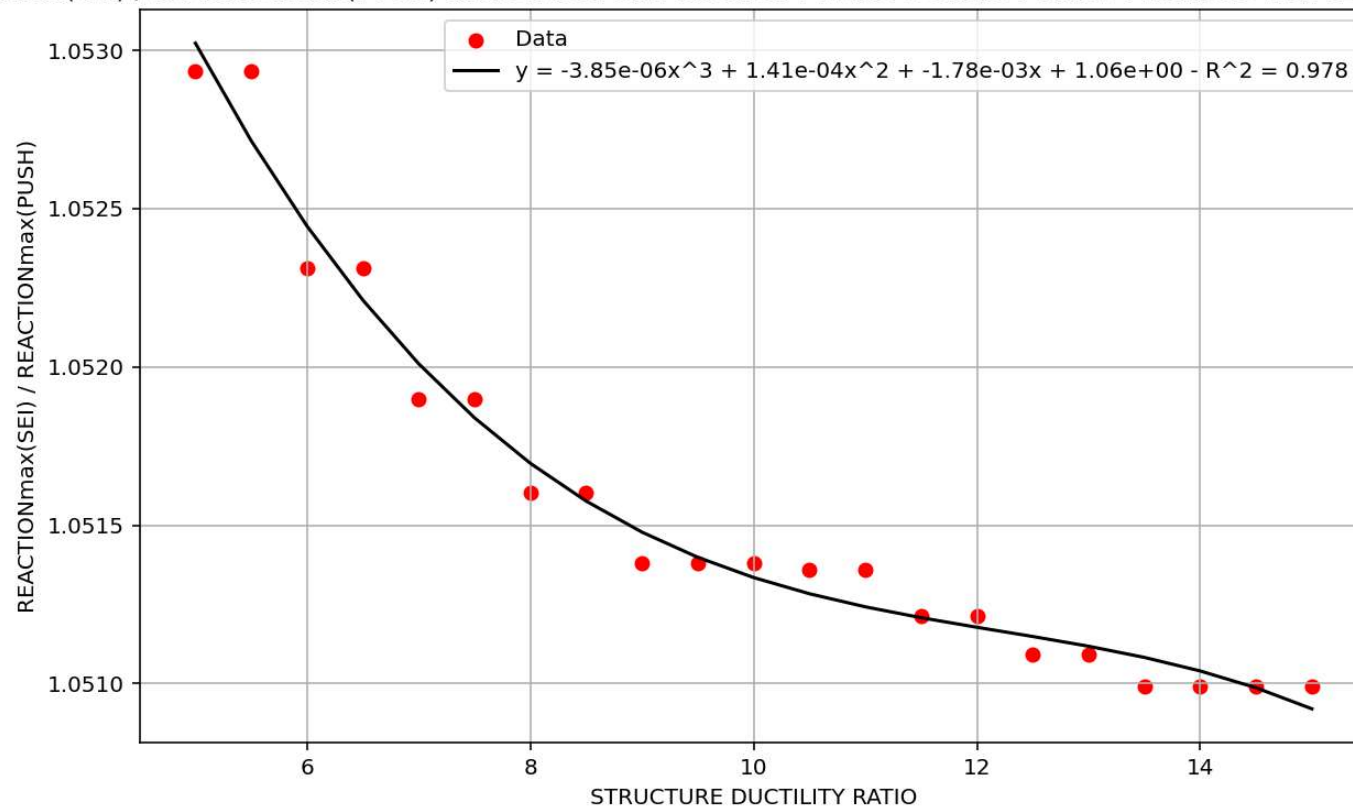
MAX. PERIOD DURING DYN. ANAL. [s] and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



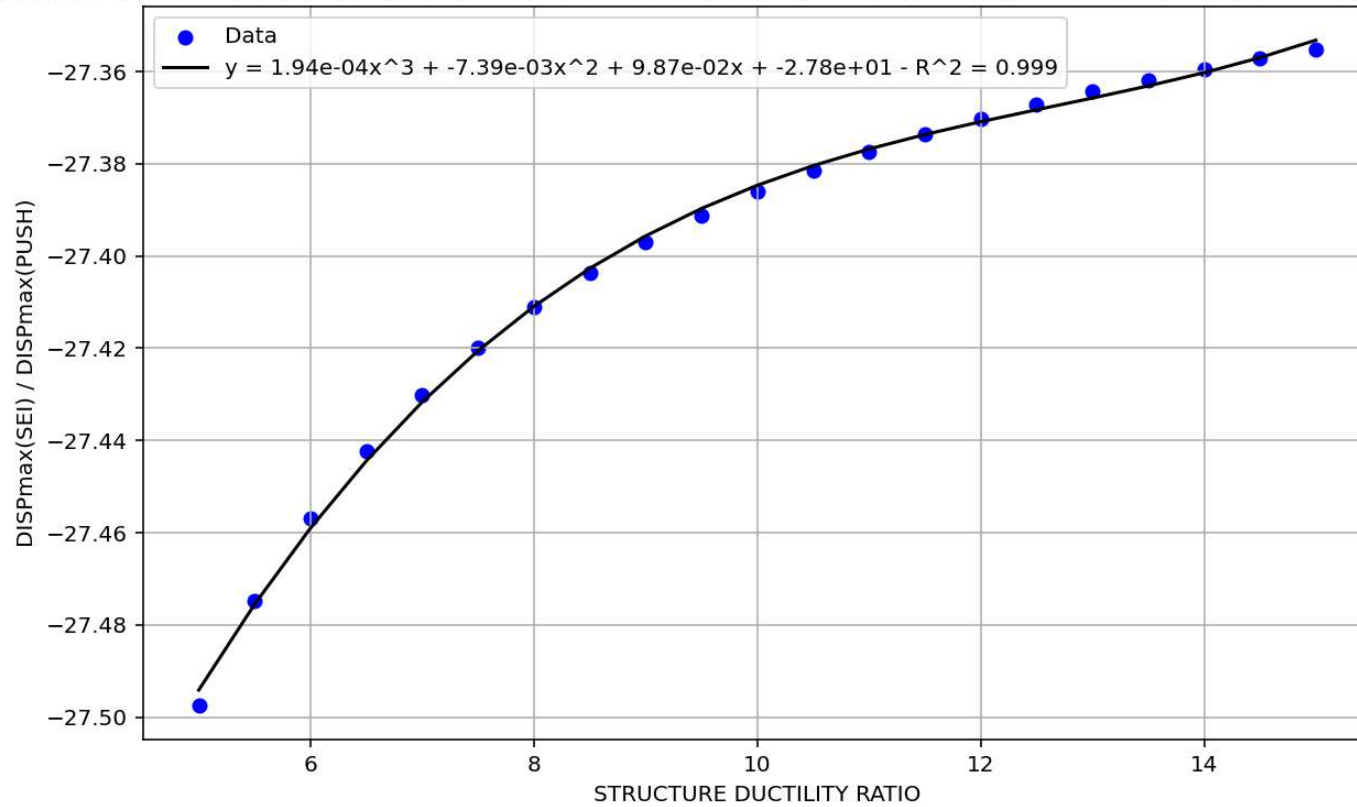
Tmax(SEI) / Tmax(PUSH) and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



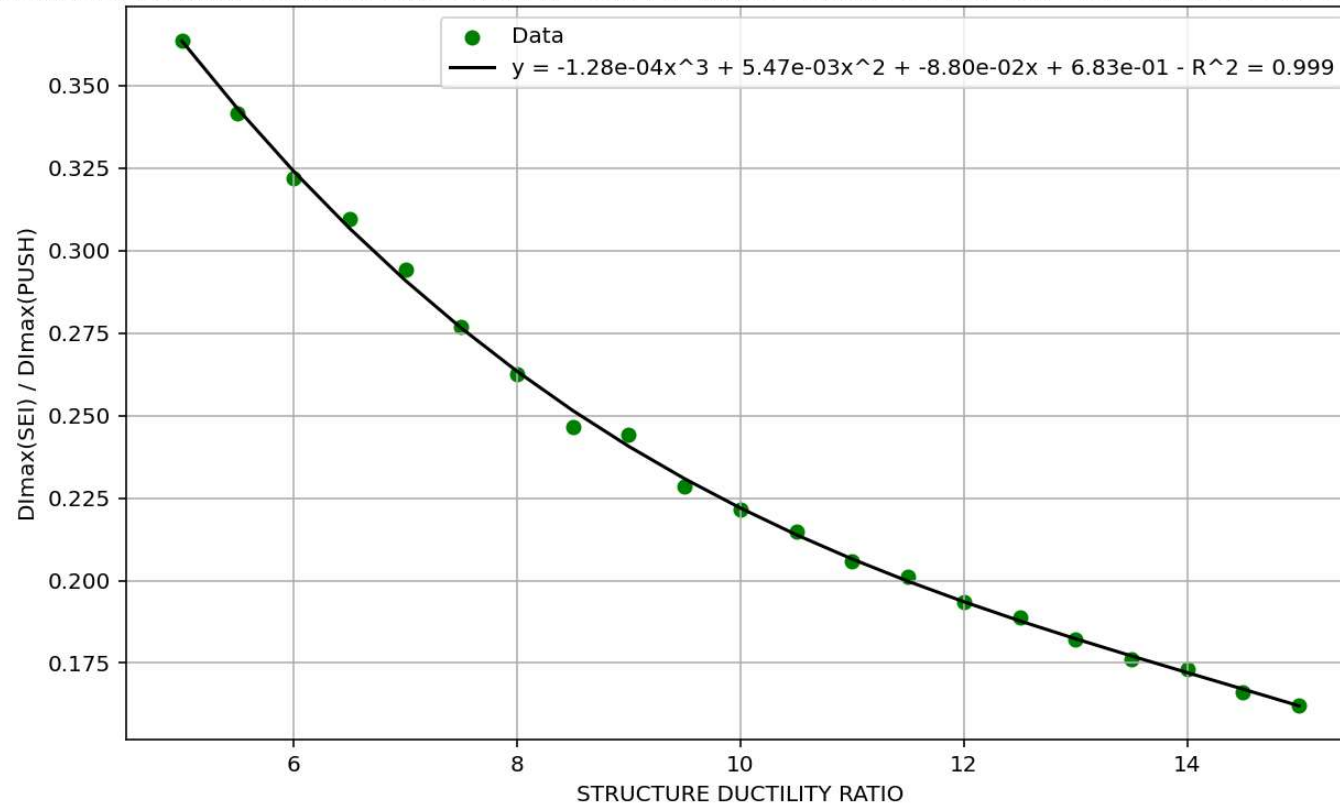
REACTIONmax(SEI) / REACTIONmax(PUSH) and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



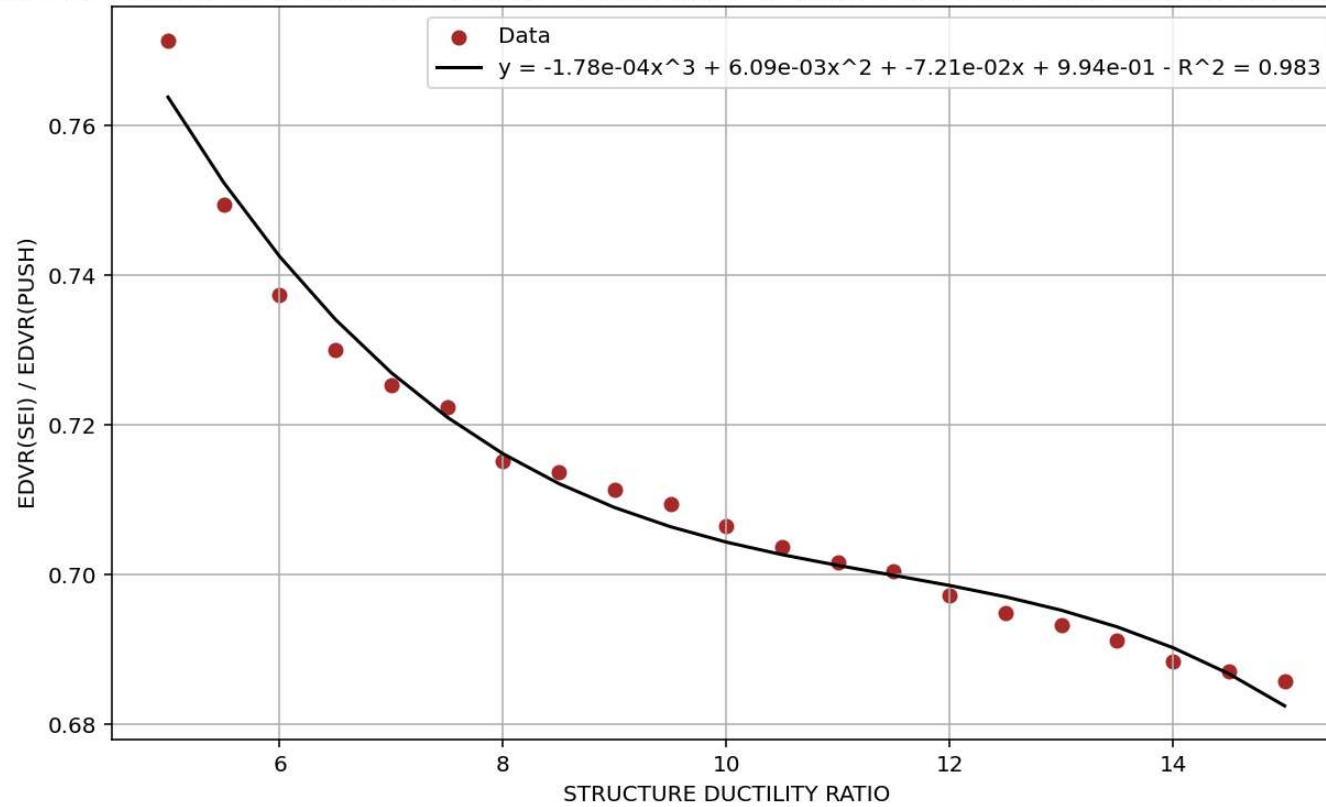
DISPmax(SEI) / DISPmax(PUSH) and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS



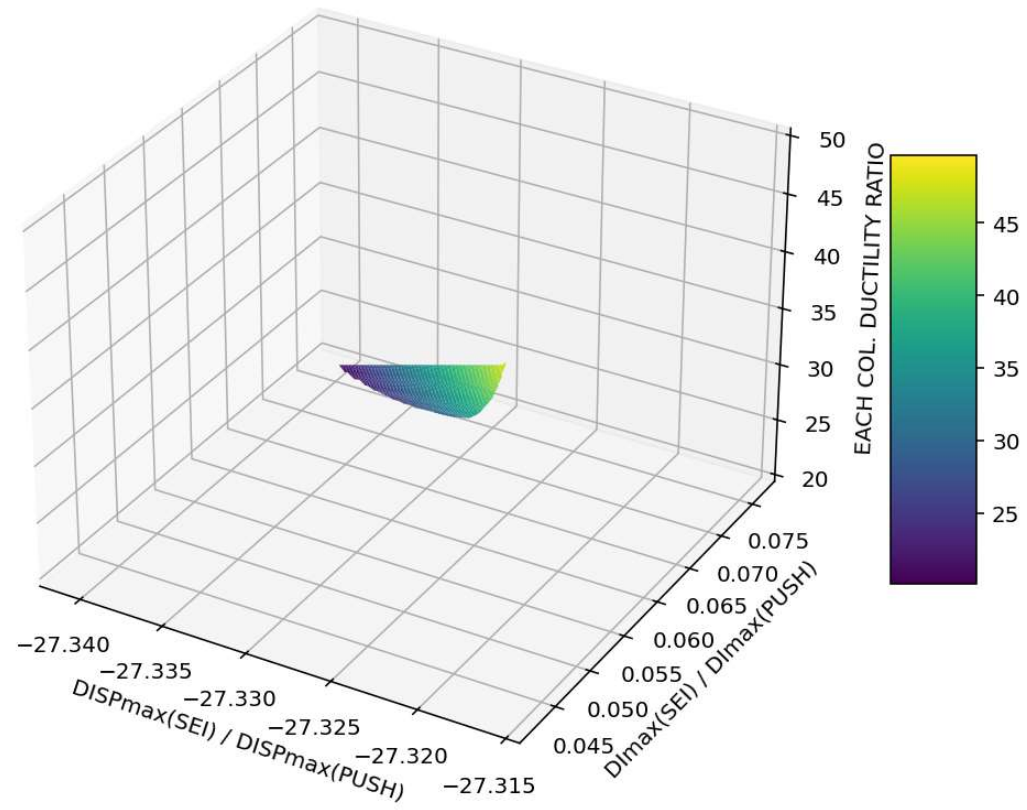
D_{lmax}(SEI) / D_{lmax}(PUSH) and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS

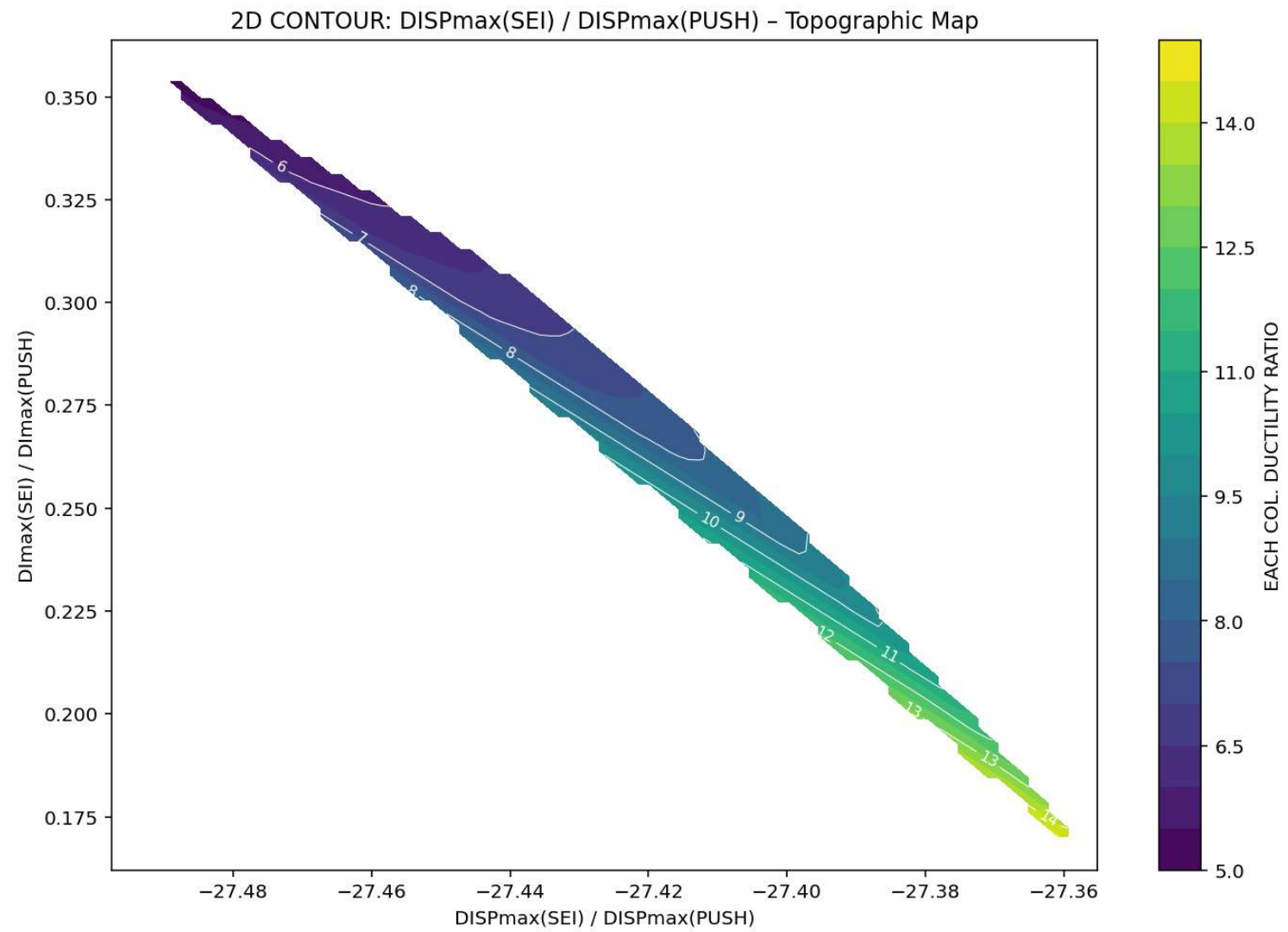


EDVR(SEI) / EDVR(PUSH) and STRUCTURE DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS

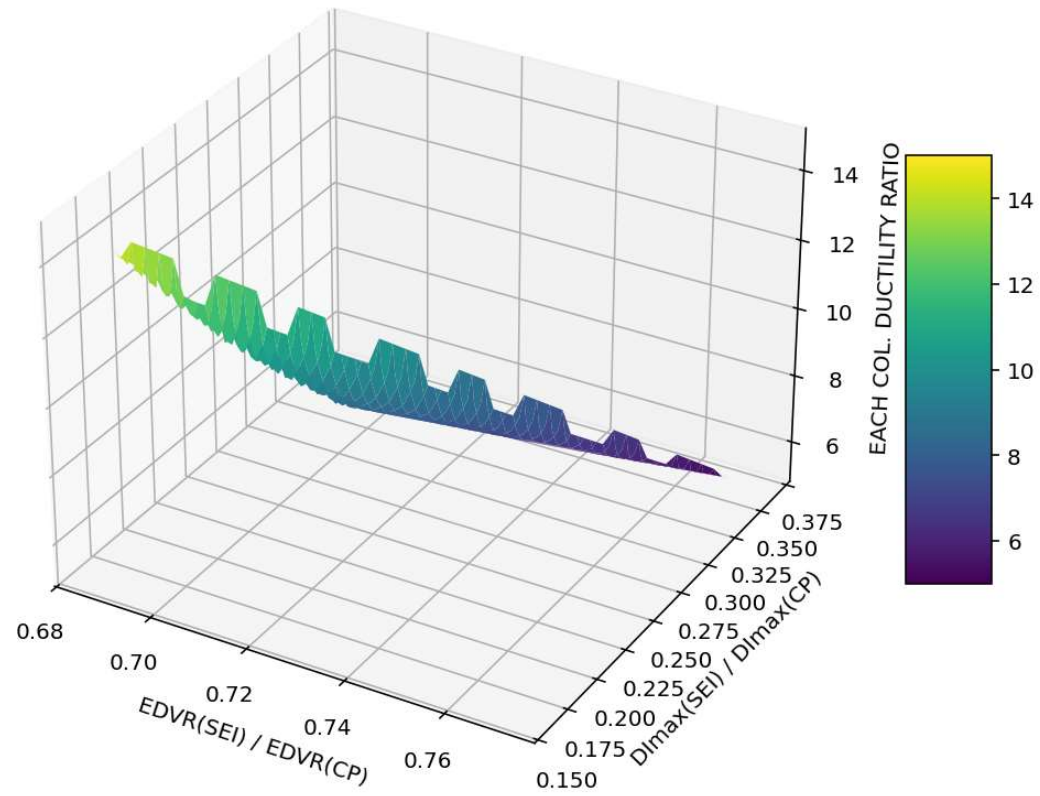


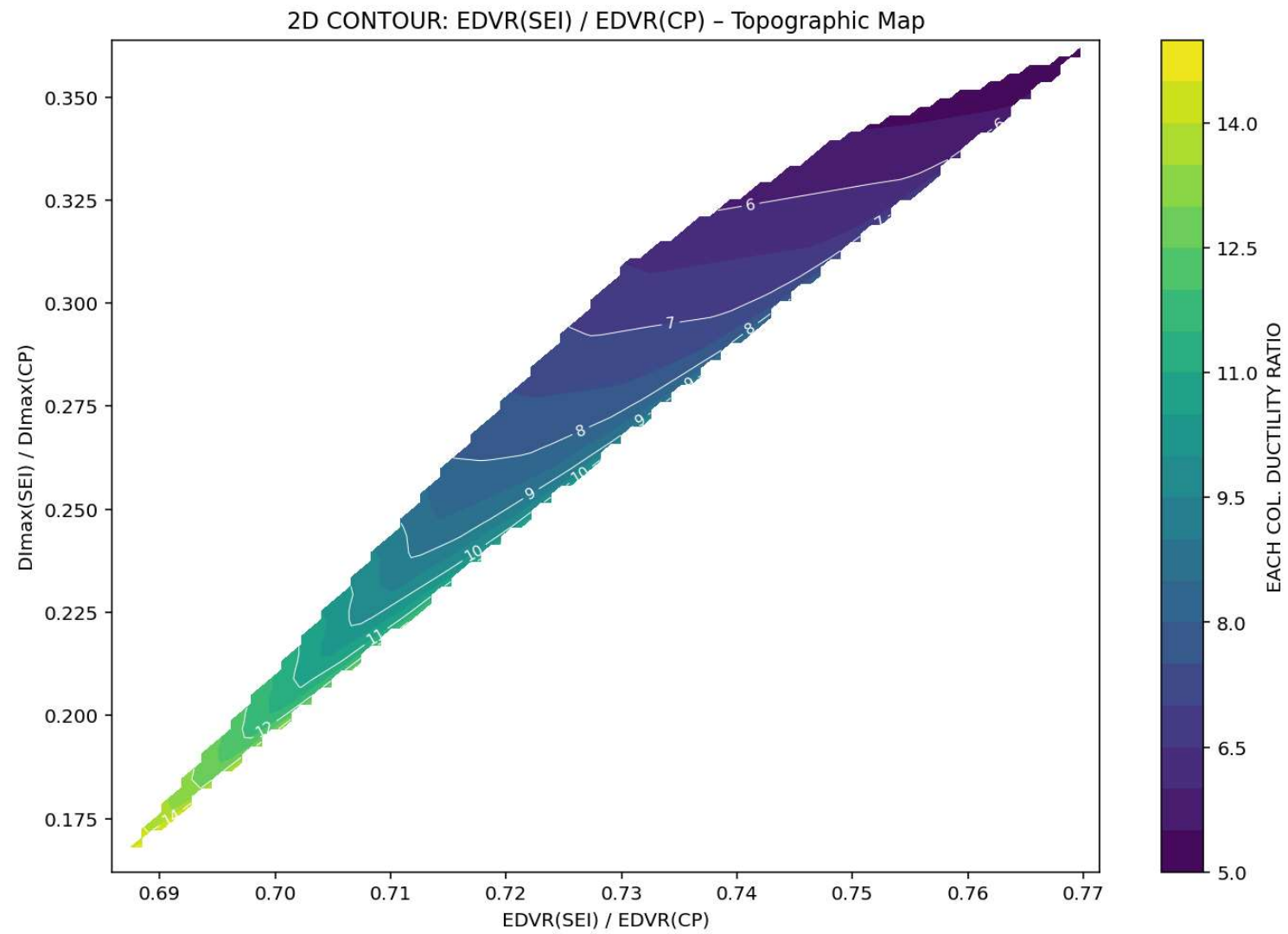
3D CONTOUR: $\text{DISPmax(SEI)} / \text{DISPmax(PUSH)}$ - $\text{Dlmax(SEI)} / \text{Dlmax(PUSH)}$ - EACH COL. DUCTILITY RATIO



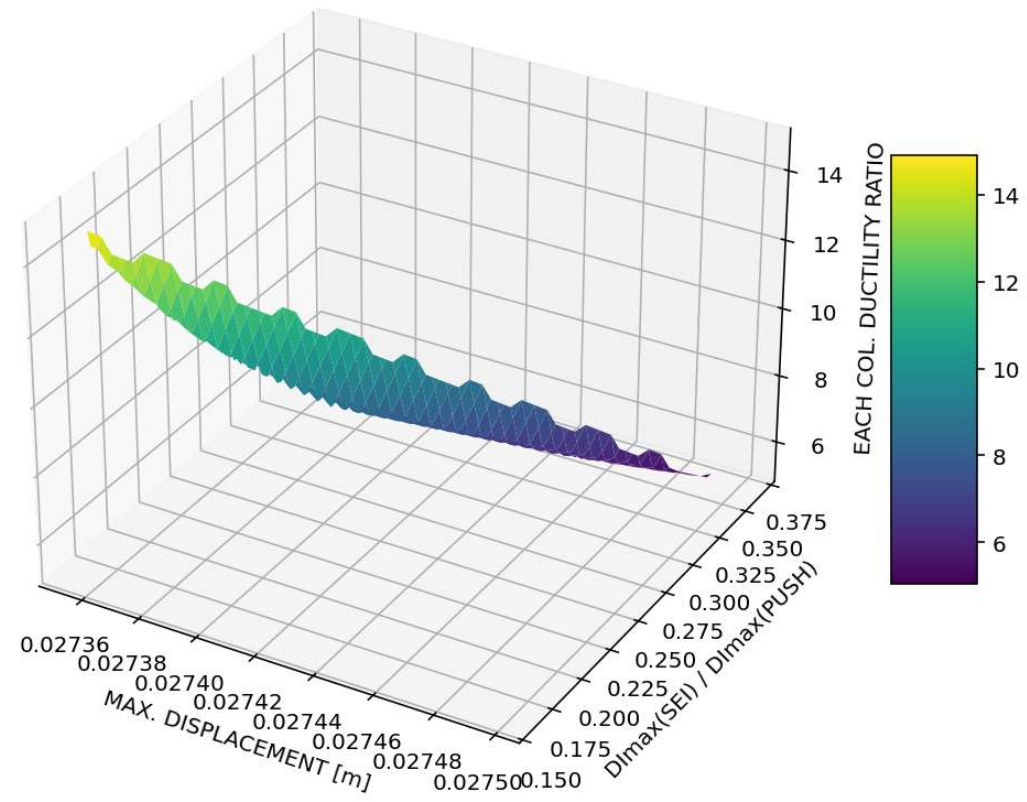


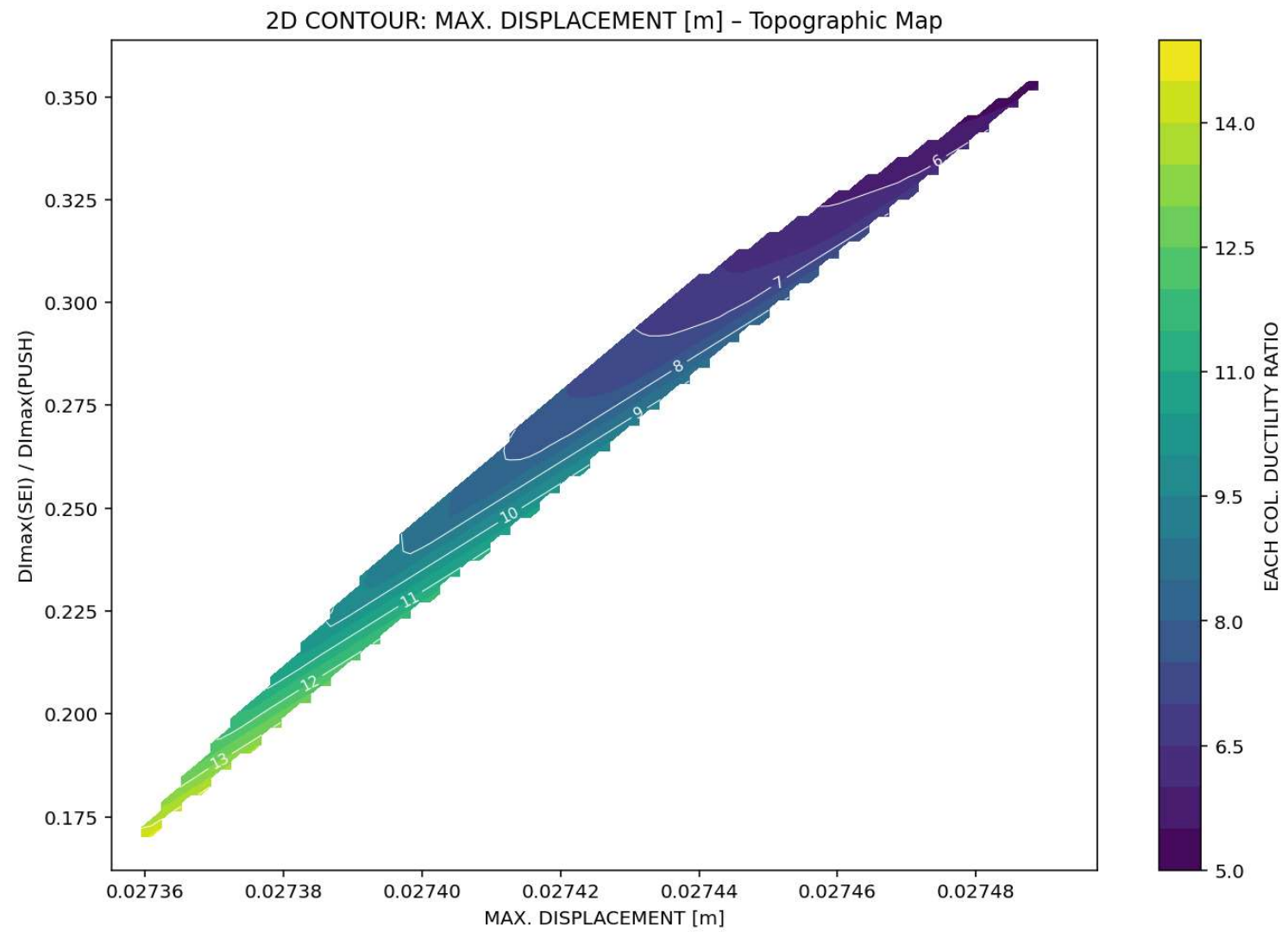
3D CONTOUR: $EDVR(SEI) / EDVR(CP) - D_{max}(SEI) / D_{max}(CP) -$ EACH COL. DUCTILITY RATIO



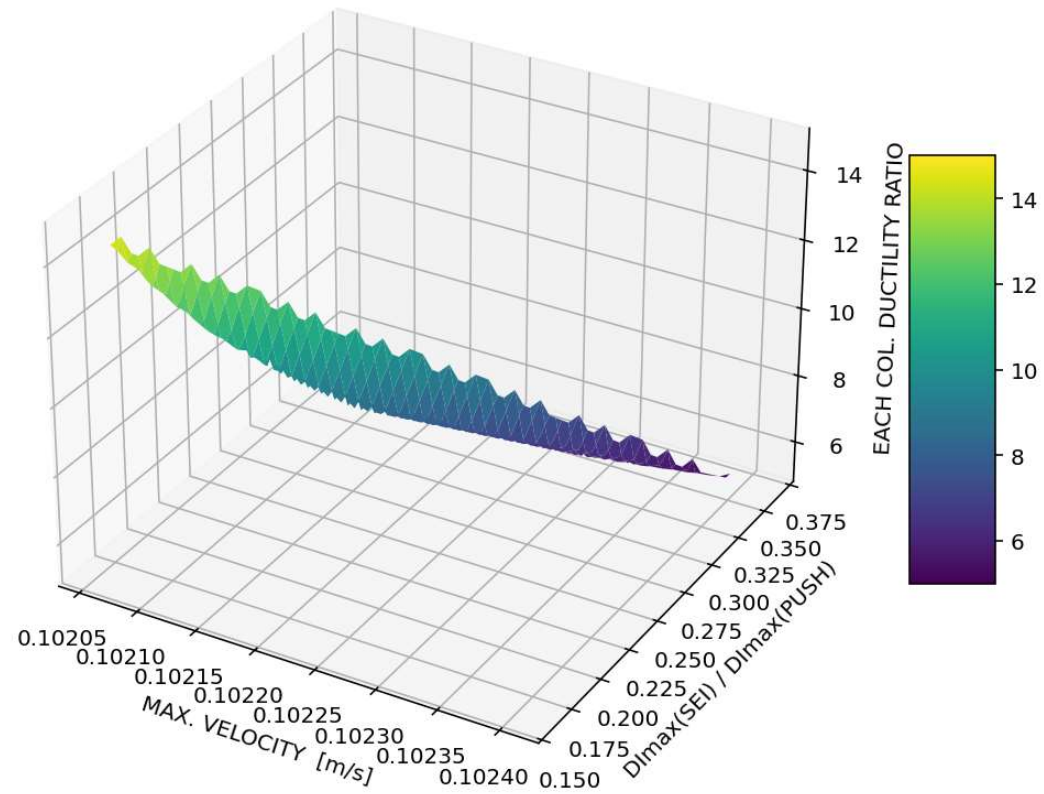


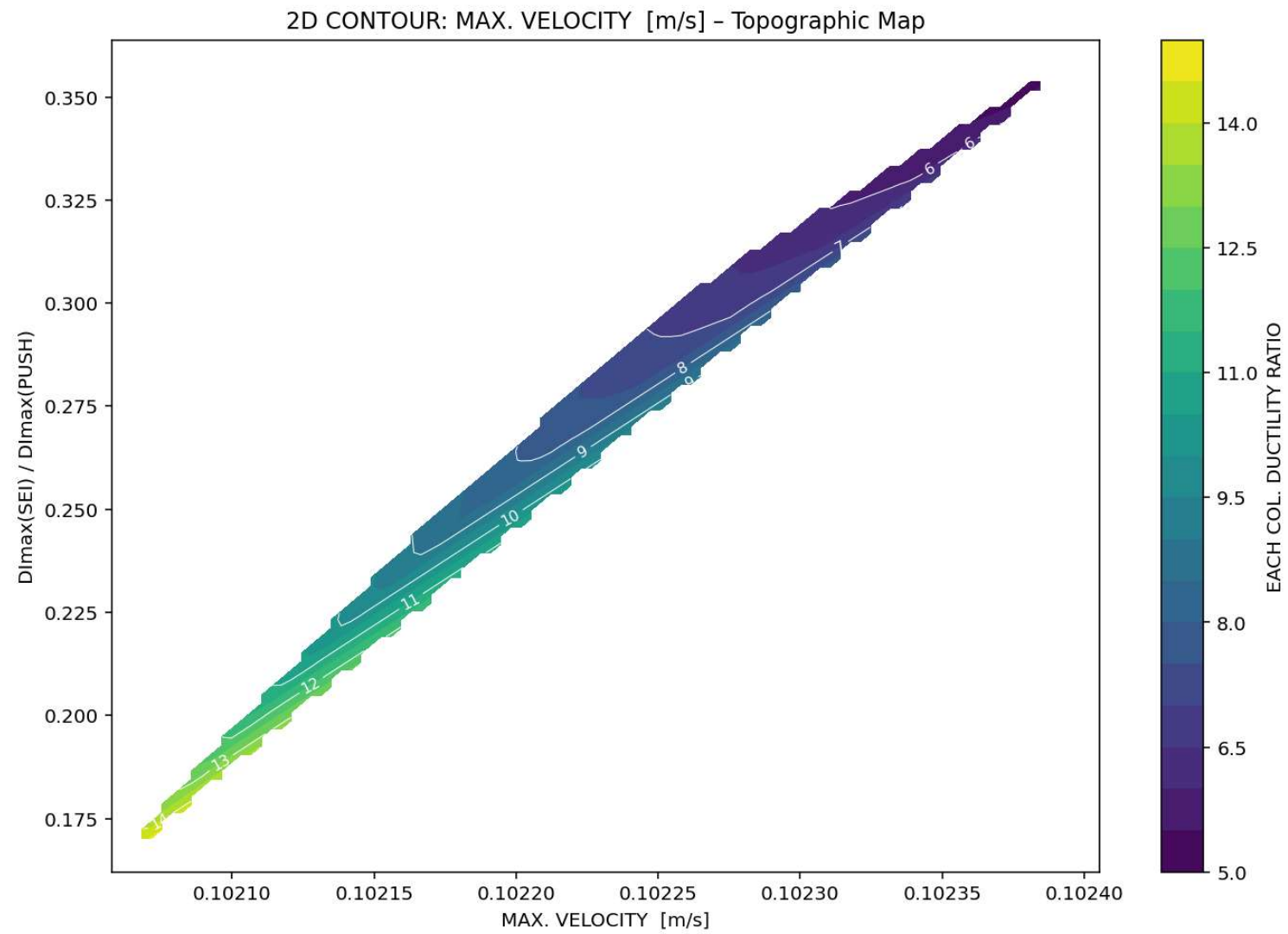
3D CONTOUR: MAX. DISPLACEMENT [m] - $D_{\max}(\text{SEI}) / D_{\max}(\text{PUSH})$ - EACH COL. DUCTILITY RATIO



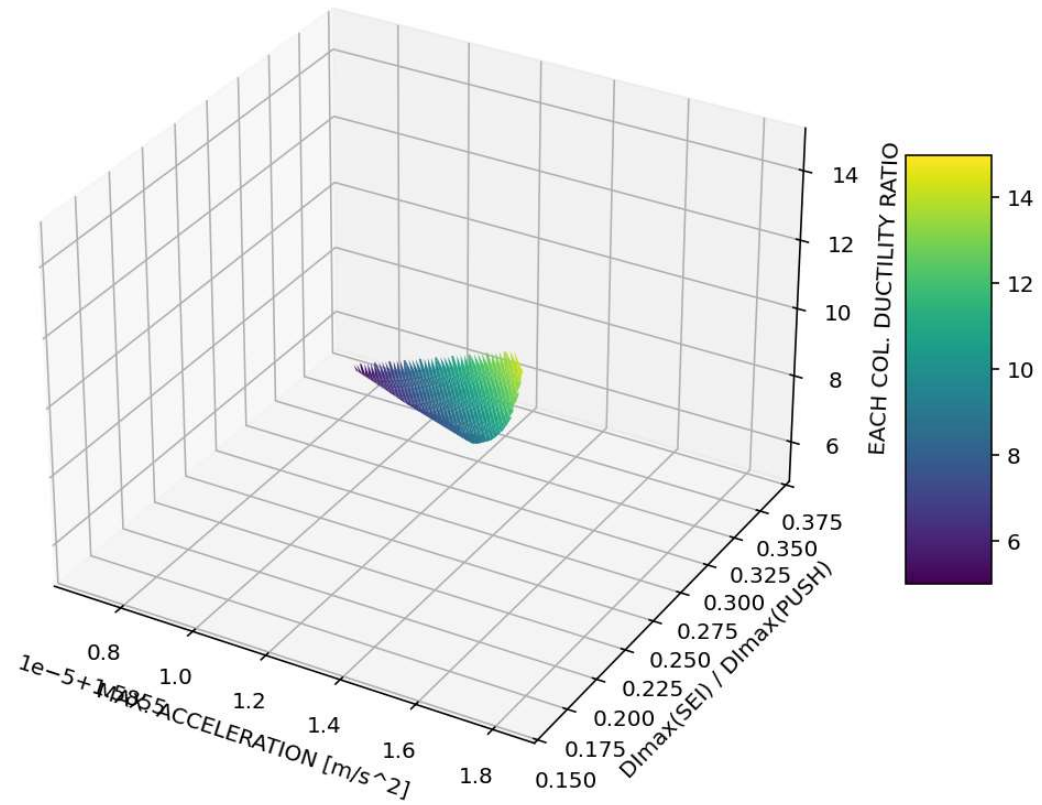


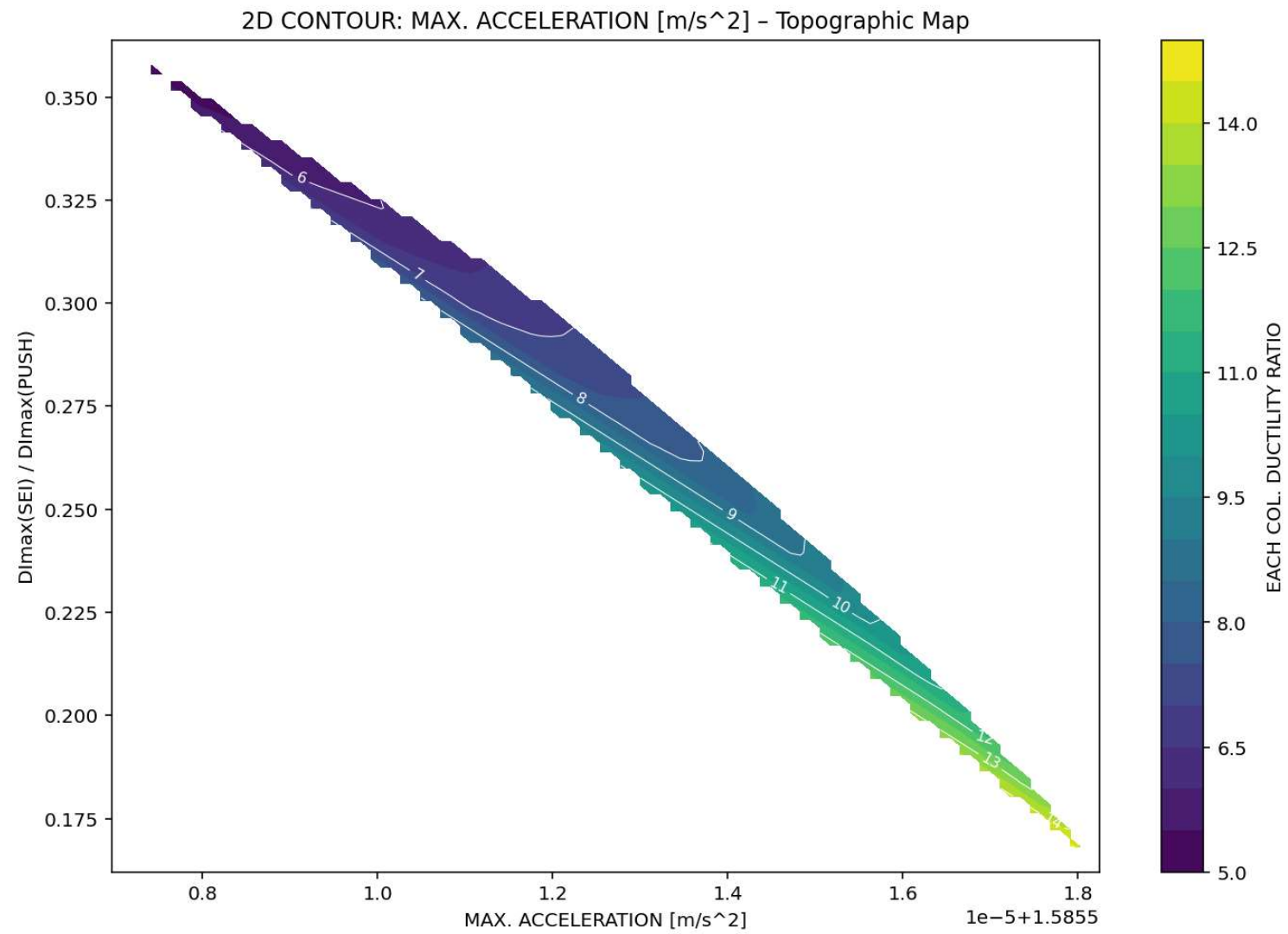
3D CONTOUR: MAX. VELOCITY [m/s] - $D_{\max}(\text{SEI}) / D_{\max}(\text{PUSH})$ - EACH COL. DUCTILITY RATIO

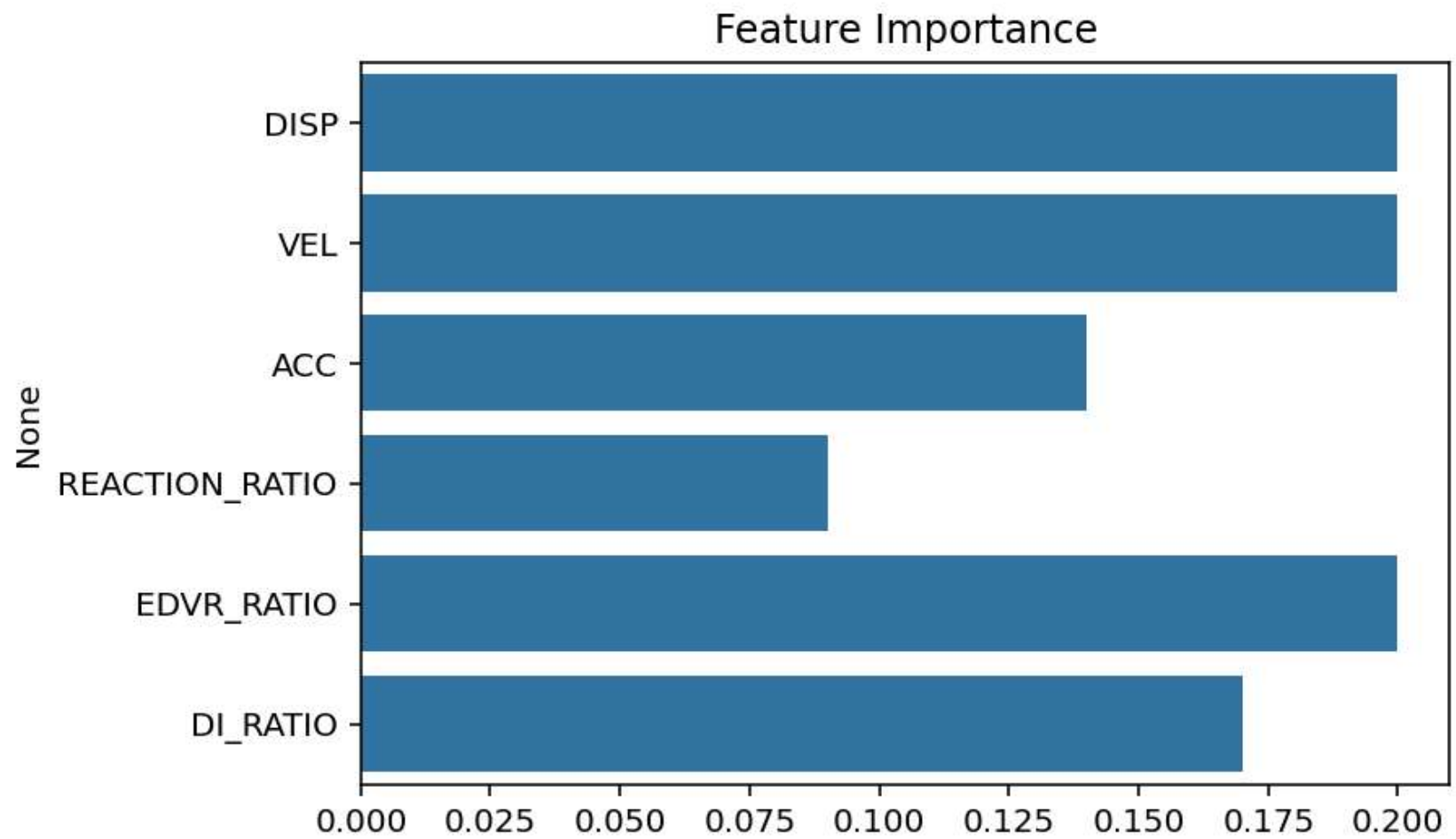


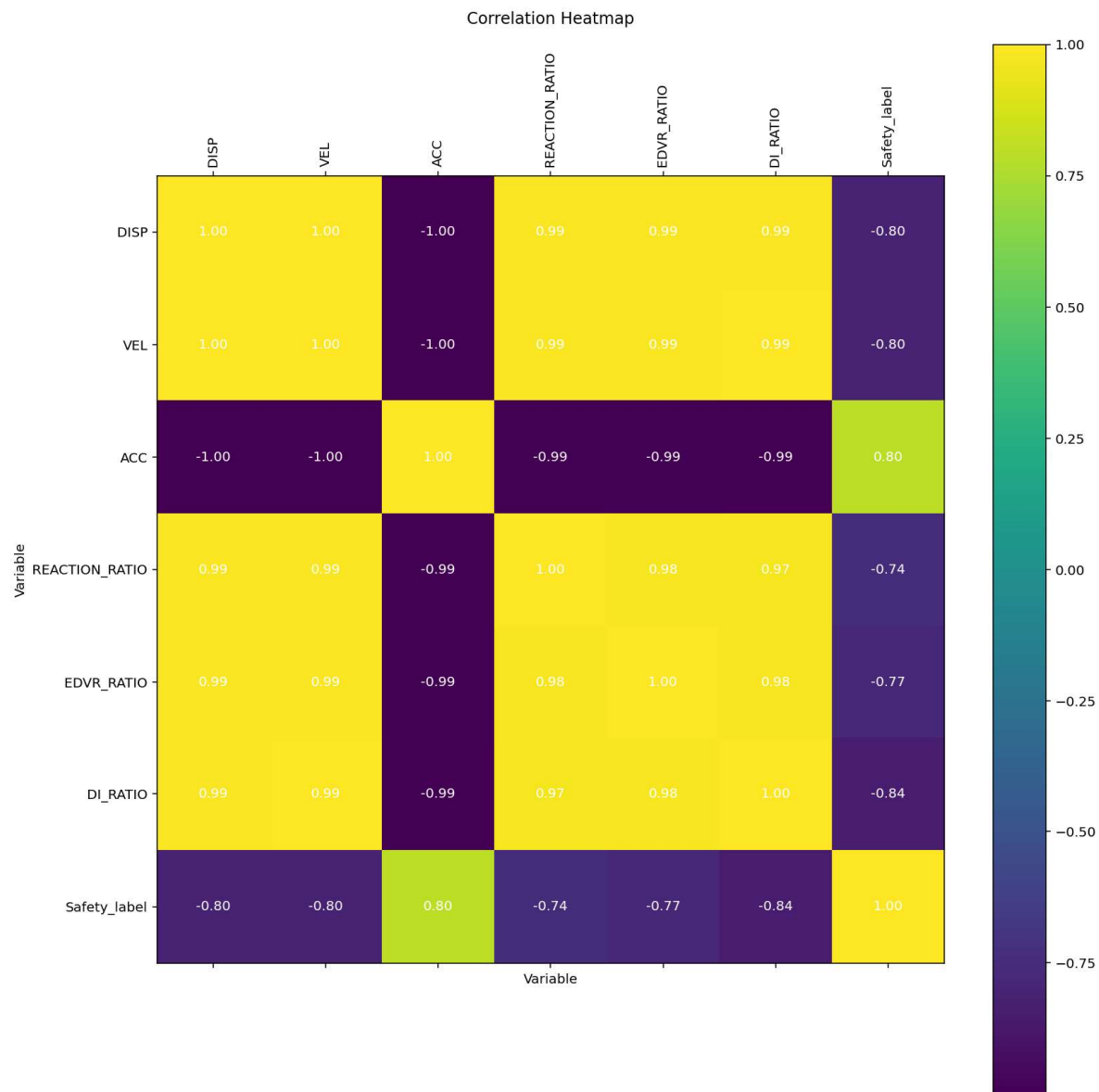


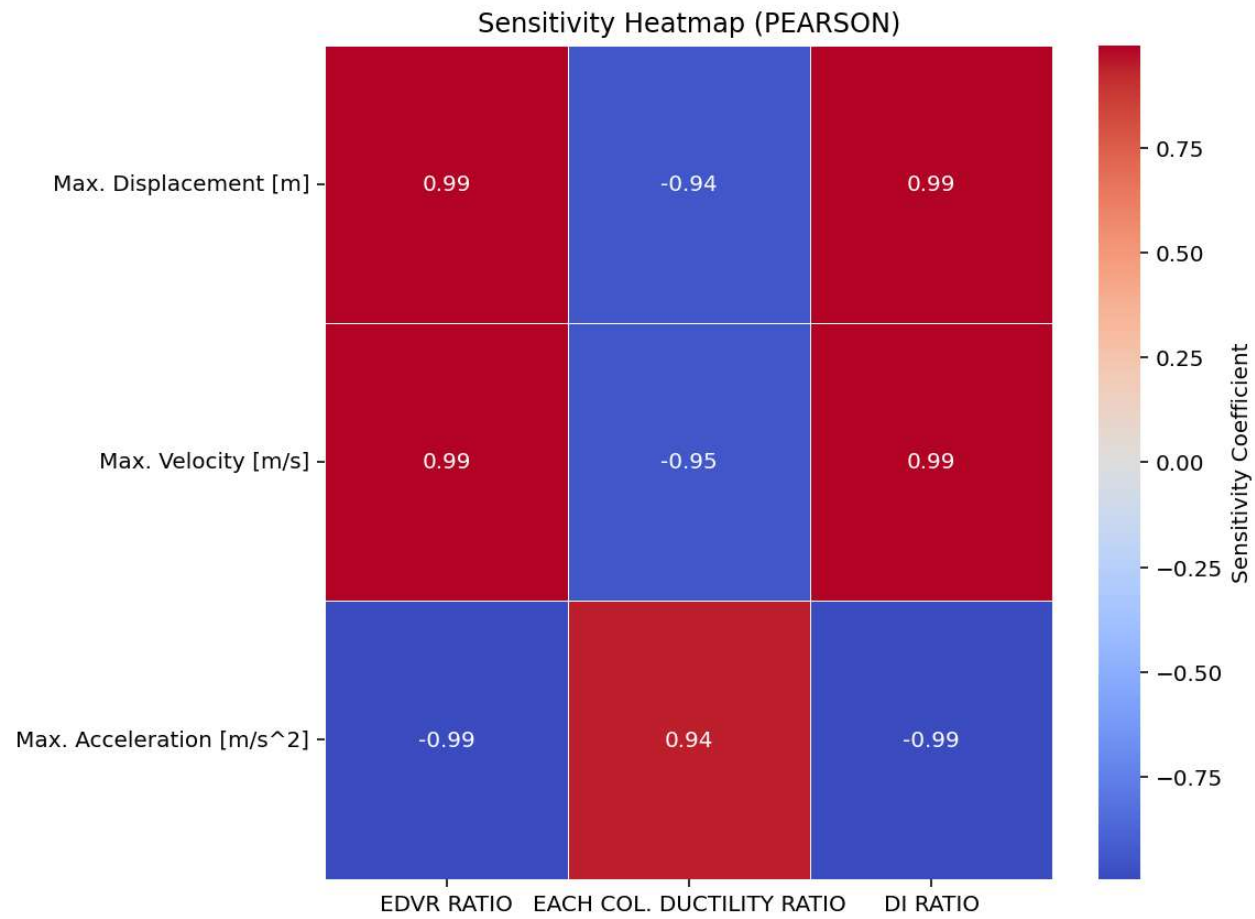
3D CONTOUR: MAX. ACCELERATION [m/s²] - D_{lmax}(SEI) / D_{lmax}(PUSH) - EACH COL. DUCTILITY RATIO

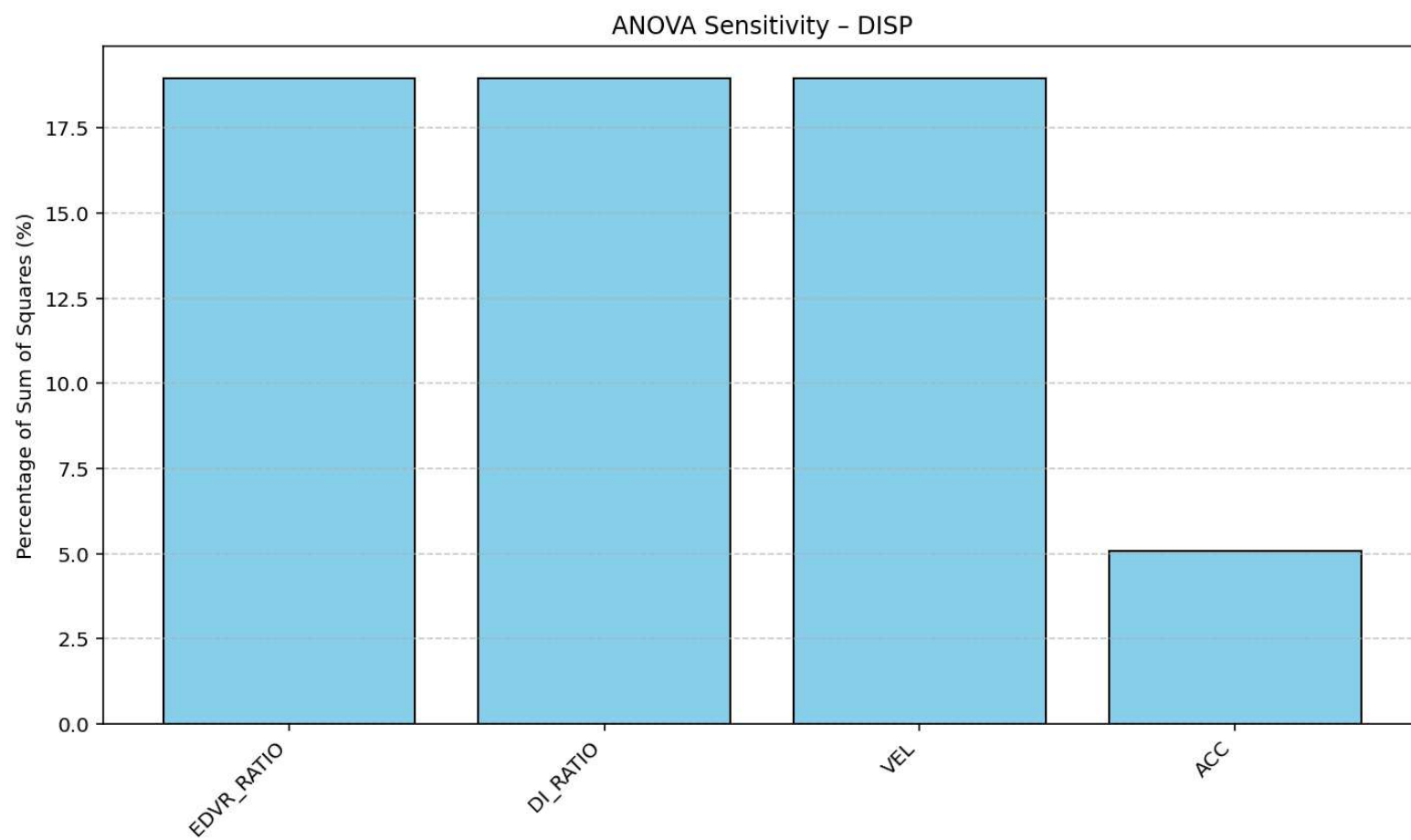












ANOVA Sensitivity - DISP

